

# rph8-g15q: formula evidence

The page, the confidence and the picture of an inline formula are its HOST LINE’s — a formula has none of its own. A line’s confidence is not a formula’s.

## Inline formulas (first occurrence)

| Identifier       | Page | Conf.   | LaTeX source                                               | Rendered                                                   | Image                                                                                                                          |
|------------------|------|---------|------------------------------------------------------------|------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| rph8-g15q_F00001 | 1    | ■ 0.280 | $\{ \}^{\{\text{@}\}}$                                     | @                                                          | Arthur J. Parzygnat <sup>1,2,*</sup> Tai-Danae Bradley <sup>3,4,†</sup> Andrew Vlasic <sup>5,‡</sup> and Anh Pham <sup>6</sup> |
| rph8-g15q_F00002 | 1    | ■ 0.280 | $\{ \}^{\{1,2,*\}}$                                        | 1,2,*                                                      | Arthur J. Parzygnat <sup>1,2,*</sup> Tai-Danae Bradley <sup>3,4,†</sup> Andrew Vlasic <sup>5,‡</sup> and Anh Pham <sup>6</sup> |
| rph8-g15q_F00003 | 1    | ■ 0.280 | $\{ \}^{\{3,4,\dag\}}$                                     | 3,4,†                                                      | Arthur J. Parzygnat <sup>1,2,*</sup> Tai-Danae Bradley <sup>3,4,†</sup> Andrew Vlasic <sup>5,‡</sup> and Anh Pham <sup>6</sup> |
| rph8-g15q_F00004 | 1    | ■ 0.280 | $\{ \}^{\{5,\ddagger\}}$                                   | 5,‡                                                        | Arthur J. Parzygnat <sup>1,2,*</sup> Tai-Danae Bradley <sup>3,4,†</sup> Andrew Vlasic <sup>5,‡</sup> and Anh Pham <sup>6</sup> |
| rph8-g15q_F00005 | 1    | ■ 1.000 | $\{ \}^{\{\dag\}}$                                         | †                                                          | <sup>†</sup> Contact author: tai.danae@math3ma.com                                                                             |
| rph8-g15q_F00006 | 1    | ■ 1.000 | $\{ \}^{\{\ddagger\}}$                                     | ‡                                                          | <sup>‡</sup> Contact author: avlasic@deloitte.com                                                                              |
| rph8-g15q_F00007 | 3    | ■ 1.000 | $x$                                                        | $x$                                                        | transfers classical data $x$ onto the quantum computer, the variational                                                        |
| rph8-g15q_F00008 | 3    | ■ 1.000 | $\theta$                                                   | $\theta$                                                   | block whose parameters $\theta$ can be modified so as to optimize some                                                         |
| rph8-g15q_F00009 | 3    | ■ 0.466 | $\rangle$                                                  | $\rangle$                                                  | 1011 is assigned to the four-qubit state $ 1011\rangle$ . Meanwhile,                                                           |
| rph8-g15q_F00010 | 3    | ■ 1.000 | $v=\text{left}(v_{\{1\}}, \ldots, v_{\{d\}}\text{right})$  | $v = (v_1, \dots, v_d)$                                    | encoding vectors $v = (v_1, \dots, v_d)$ in $\mathbb{R}^d$ into quantum states                                                 |
| rph8-g15q_F00011 | 3    | ■ 1.000 | $\mathbb{R}^d$                                             | $\mathbb{R}^d$                                             | encoding vectors $v = (v_1, \dots, v_d)$ in $\mathbb{R}^d$ into quantum states                                                 |
| rph8-g15q_F00012 | 3    | ■ 1.000 | $\mathcal{H}$                                              | $\mathcal{H}$                                              | [8,60,61]. Given a Hilbert space $\mathcal{H}$ , let $\mathcal{S}(\mathcal{H})$ denote the set                                 |
| rph8-g15q_F00013 | 3    | ■ 1.000 | $\mathscr{S}(\mathcal{H})$                                 | $\mathscr{S}(\mathcal{H})$                                 | [8,60,61]. Given a Hilbert space $\mathcal{H}$ , let $\mathcal{S}(\mathcal{H})$ denote the set                                 |
| rph8-g15q_F00014 | 3    | ■ 1.000 | $\mathscr{U}(\mathcal{H})$                                 | $\mathscr{U}(\mathcal{H})$                                 | let $\mathcal{U}(\mathcal{H})$ denote the set of unitary operators on $\mathcal{H}$ . Although                                 |
| rph8-g15q_F00015 | 3    | ■ 1.000 | $\mathcal{X}$                                              | $\mathcal{X}$                                              | We will often let $\mathcal{X}$ denote a set, called a <i>data domain</i> or                                                   |
| rph8-g15q_F00016 | 3    | ■ 1.000 | $X$                                                        | $X$                                                        | $X$ typically used for the dataset itself (e.g., a sample), which                                                              |
| rph8-g15q_F00017 | 3    | ■ 1.000 | $\rho: \mathcal{X} \rightarrow \mathscr{S}(\mathcal{H})$   | $\rho: \mathcal{X} \rightarrow \mathscr{S}(\mathcal{H})$   | main $\mathcal{X}$ into a Hilbert space $\mathcal{H}$ is a function $\rho: \mathcal{X} \rightarrow \mathscr{S}(\mathcal{H})$ . |
| rph8-g15q_F00018 | 3    | ■ 1.000 | $U: \mathcal{X} \rightarrow \mathscr{U}(\mathcal{H})$      | $U: \mathcal{X} \rightarrow \mathscr{U}(\mathcal{H})$      | A <i>quantum unitary encoding</i> is a function $U: \mathcal{X} \rightarrow \mathscr{U}(\mathcal{H})$ .                        |
| rph8-g15q_F00019 | 3    | ■ 1.000 | $ \psi_0\rangle \in \mathcal{H}$                           | $ \psi_0\rangle \in \mathcal{H}$                           | $ \psi_0\rangle \in \mathcal{H}$ , gives rise to a quantum state encoding $\rho: \mathcal{X} \rightarrow$                      |
| rph8-g15q_F00020 | 3    | ■ 1.000 | $\rho: \mathcal{X} \rightarrow$                            | $\rho: \mathcal{X} \rightarrow$                            | $ \psi_0\rangle \in \mathcal{H}$ , gives rise to a quantum state encoding $\rho: \mathcal{X} \rightarrow$                      |
| rph8-g15q_F00021 | 4    | ■ 1.000 | $G$                                                        | $G$                                                        | are equipped with an action of a group $G$ , and a quantum state                                                               |
| rph8-g15q_F00022 | 4    | ■ 1.000 | $\alpha: G \rightarrow \text{Aut}(\mathcal{X})$            | $\alpha: G \rightarrow \text{Aut}(\mathcal{X})$            | homomorphism $\alpha: G \rightarrow \text{Aut}(\mathcal{X})$ , where $\text{Aut}(\mathcal{X})$ denotes the                     |
| rph8-g15q_F00023 | 4    | ■ 1.000 | $\text{Aut}(\mathcal{X})$                                  | $\text{Aut}(\mathcal{X})$                                  | homomorphism $\alpha: G \rightarrow \text{Aut}(\mathcal{X})$ , where $\text{Aut}(\mathcal{X})$ denotes the                     |
| rph8-g15q_F00024 | 4    | ■ 1.000 | $\alpha$                                                   | $\alpha$                                                   | map $\alpha$ is referred to as the <i>group action</i> . Such a $G$ -set $\mathcal{X}$ is                                      |
| rph8-g15q_F00025 | 4    | ■ 1.000 | $(\mathcal{X}, \alpha)$                                    | $(\mathcal{X}, \alpha)$                                    | often written as a pair $(\mathcal{X}, \alpha)$ .                                                                              |
| rph8-g15q_F00026 | 4    | ■ 1.000 | $g \in G$                                                  | $g \in G$                                                  | with a group element $g \in G$ is often written as $\alpha_g: \mathcal{X} \rightarrow \mathcal{X}$ .                           |
| rph8-g15q_F00027 | 4    | ■ 1.000 | $\alpha_g: \mathcal{X} \rightarrow \mathcal{X}$            | $\alpha_g: \mathcal{X} \rightarrow \mathcal{X}$            | with a group element $g \in G$ is often written as $\alpha_g: \mathcal{X} \rightarrow \mathcal{X}$ .                           |
| rph8-g15q_F00028 | 4    | ■ 1.000 | $x \in \mathcal{X}$                                        | $x \in \mathcal{X}$                                        | Its action on an element $x \in \mathcal{X}$ is written as $\alpha_g(x)$ , which is                                            |
| rph8-g15q_F00029 | 4    | ■ 1.000 | $\alpha_g(x)$                                              | $\alpha_g(x)$                                              | Its action on an element $x \in \mathcal{X}$ is written as $\alpha_g(x)$ , which is                                            |
| rph8-g15q_F00030 | 4    | ■ 1.000 | $\alpha_{gh}(x) = \alpha_g(\alpha_h(x))$                   | $\alpha_{gh}(x) = \alpha_g(\alpha_h(x))$                   | in that $\alpha_{gh}(x) = \alpha_g(\alpha_h(x))$ for all $g, h \in G$ and $x \in \mathcal{X}$ as well                          |
| rph8-g15q_F00031 | 4    | ■ 1.000 | $g, h \in G$                                               | $g, h \in G$                                               | in that $\alpha_{gh}(x) = \alpha_g(\alpha_h(x))$ for all $g, h \in G$ and $x \in \mathcal{X}$ as well                          |
| rph8-g15q_F00032 | 4    | ■ 0.987 | $\alpha_{1_G} = \text{id}_{\mathcal{X}}$                   | $\alpha_{1_G} = \text{id}_{\mathcal{X}}$                   | as $\alpha_{1_G} = \text{id}_{\mathcal{X}}$ , where $1_G$ is the identity in the group $G$ .                                   |
| rph8-g15q_F00033 | 4    | ■ 0.987 | $1_G$                                                      | $1_G$                                                      | as $\alpha_{1_G} = \text{id}_{\mathcal{X}}$ , where $1_G$ is the identity in the group $G$ .                                   |
| rph8-g15q_F00034 | 4    | ■ 1.000 | $V: G \rightarrow$                                         | $V: G \rightarrow$                                         | As an example, if $\mathcal{H}$ is a Hilbert space, and if $V: G \rightarrow$                                                  |
| rph8-g15q_F00035 | 4    | ■ 1.000 | $\text{Aut}(\mathcal{H})$                                  | $\text{Aut}(\mathcal{H})$                                  | viewed as a subset of $\text{Aut}(\mathcal{H})$ , since unitary operators can be                                               |
| rph8-g15q_F00036 | 4    | ■ 1.000 | $\text{Ad}_{V_g} \in \text{Aut}(\mathscr{S}(\mathcal{H}))$ | $\text{Ad}_{V_g} \in \text{Aut}(\mathscr{S}(\mathcal{H}))$ | operator $\text{Ad}_{V_g} \in \text{Aut}(\mathscr{S}(\mathcal{H}))$ defined by                                                 |
| rph8-g15q_F00037 | 4    | ■ 1.000 | $\alpha_{(1,0)}$                                           | $\alpha_{(1,0)}$                                           | binary classification task with discrete symmetries given by $\alpha_{(1,0)}$ ,                                                |
| rph8-g15q_F00038 | 4    | ■ 1.000 | $x_2 = x_1, \alpha_{(0,1)}$                                | $x_2 = x_1, \alpha_{(0,1)}$                                | a reflection along the line $x_2 = x_1$ , $\alpha_{(0,1)}$ , an inversion, and $\alpha_{(1,1)}$ ,                              |
| rph8-g15q_F00039 | 4    | ■ 1.000 | $\alpha_{(1,1)}$                                           | $\alpha_{(1,1)}$                                           | a reflection along the line $x_2 = x_1$ , $\alpha_{(0,1)}$ , an inversion, and $\alpha_{(1,1)}$ ,                              |
| rph8-g15q_F00040 | 4    | ■ 1.000 | $x_2 = -x_1$                                               | $x_2 = -x_1$                                               | a reflection along the line $x_2 = -x_1$ . The structure is periodic so                                                        |
| rph8-g15q_F00041 | 4    | ■ 1.000 | $\sigma \in \mathscr{S}(\mathcal{H})$                      | $\sigma \in \mathscr{S}(\mathcal{H})$                      | for all $\sigma \in \mathscr{S}(\mathcal{H})$ . Here, “Ad” stands for the <i>adjoint</i> action.                               |
| rph8-g15q_F00042 | 4    | ■ 0.987 | $(\mathscr{S}(\mathcal{H}), \text{Ad}_V)$                  | $(\mathscr{S}(\mathcal{H}), \text{Ad}_V)$                  | Thus, $(\mathscr{S}(\mathcal{H}), \text{Ad}_V)$ is a $G$ -set as well.                                                         |

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|------------------|------|---------|--------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| rph8-g15q_F00043 | 4    | ■ 1.000 | $V: G \rightarrow \mathcal{U}(\mathcal{H})$                                          | $V: G \rightarrow \mathcal{U}(\mathcal{H})$                                          | tation $V: G \rightarrow \mathcal{U}(\mathcal{H})$ of $G$ on the Hilbert space $\mathcal{H}$ such that                                          |
| rph8-g15q_F00044 | 4    | ■ 1.000 | $\rho$                                                                               | $\rho$                                                                               | $\rho$ is $G$ -equivariant, that is,                                                                                                            |
| rph8-g15q_F00045 | 4    | ■ 0.616 | $\mathcal{X}, \alpha$                                                                | $\mathcal{X}, \alpha$                                                                | <i>Definition 4.</i> Let $(\mathcal{X}, \alpha)$ and $(\mathcal{Y}, \beta)$ be two $G$ -sets. A $G$ -                                           |
| rph8-g15q_F00046 | 4    | ■ 0.616 | $\mathcal{Y}, \beta$                                                                 | $\mathcal{Y}, \beta$                                                                 | <i>Definition 4.</i> Let $(\mathcal{X}, \alpha)$ and $(\mathcal{Y}, \beta)$ be two $G$ -sets. A $G$ -                                           |
| rph8-g15q_F00047 | 4    | ■ 0.999 | $(\mathcal{Y}, \beta)$                                                               | $(\mathcal{Y}, \beta)$                                                               | equivariant map from $(\mathcal{X}, \alpha)$ to $(\mathcal{Y}, \beta)$ is a function $f: \mathcal{X} \rightarrow$                               |
| rph8-g15q_F00048 | 4    | ■ 0.999 | $f: \mathcal{X} \rightarrow$                                                         | $f: \mathcal{X} \rightarrow$                                                         | equivariant map from $(\mathcal{X}, \alpha)$ to $(\mathcal{Y}, \beta)$ is a function $f: \mathcal{X} \rightarrow$                               |
| rph8-g15q_F00049 | 4    | ■ 1.000 | $\mathcal{Y}$                                                                        | $\mathcal{Y}$                                                                        | $\mathcal{Y}$ such that                                                                                                                         |
| rph8-g15q_F00050 | 5    | ■ 1.000 | $\mathcal{X} = \mathbb{R}^2$                                                         | $\mathcal{X} = \mathbb{R}^2$                                                         | two classes of points within a data domain $\mathcal{X} = \mathbb{R}^2$ as shown                                                                |
| rph8-g15q_F00051 | 5    | ■ 1.000 | $y: \mathcal{X} \rightarrow \mathbb{R}$                                              | $y: \mathcal{X} \rightarrow \mathbb{R}$                                              | where $y: \mathcal{X} \rightarrow \mathbb{R}$ is a function that determines the class of                                                        |
| rph8-g15q_F00052 | 5    | ■ 0.998 | $y(x) > 0$                                                                           | $y(x) > 0$                                                                           | any input data point via $y(x) > 0$ implies $x$ is of class $+1$ ,                                                                              |
| rph8-g15q_F00053 | 5    | ■ 0.516 | $y(x) < 0$                                                                           | $y(x) < 0$                                                                           | while $y(x) < 0$ implies $x$ is of class $-1$ . The symmetry de-                                                                                |
| rph8-g15q_F00054 | 5    | ■ 1.000 | $G \rightarrow \text{Aut}(\mathbb{R}^2)$                                             | $G \rightarrow \text{Aut}(\mathbb{R}^2)$                                             | $G \rightarrow \text{Aut}(\mathbb{R}^2)$ on $\mathcal{X} = \mathbb{R}^2$ given by the linear transformations                                    |
| rph8-g15q_F00055 | 5    | ■ 1.000 | $\mathcal{H} = \mathbb{C}^2 \otimes \mathbb{C}^2$                                    | $\mathcal{H} = \mathbb{C}^2 \otimes \mathbb{C}^2$                                    | Now set $\mathcal{H} = \mathbb{C}^2 \otimes \mathbb{C}^2$ and set $V: G \rightarrow \mathcal{U}(\mathcal{H})$ to be the                         |
| rph8-g15q_F00056 | 5    | ■ 0.998 | $X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \mathbb{1}_2$                     | $X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \mathbb{1}_2$                     | where $X$ is the Pauli matrix $X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ , $\mathbb{1}_2$ is the $2 \times 2$                           |
| rph8-g15q_F00057 | 5    | ■ 0.998 | $2 \times 2$                                                                         | $2 \times 2$                                                                         | where $X$ is the Pauli matrix $X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ , $\mathbb{1}_2$ is the $2 \times 2$                           |
| rph8-g15q_F00058 | 5    | ■ 0.994 | $\text{SWAP}( \psi\rangle \otimes  \phi\rangle) =  \phi\rangle \otimes  \psi\rangle$ | $\text{SWAP}( \psi\rangle \otimes  \phi\rangle) =  \phi\rangle \otimes  \psi\rangle$ | by $\text{SWAP}( \psi\rangle \otimes  \phi\rangle) =  \phi\rangle \otimes  \psi\rangle$ for all $ \psi\rangle,  \phi\rangle \in \mathbb{C}^2$ . |
| rph8-g15q_F00059 | 5    | ■ 0.994 | $ \psi\rangle,  \phi\rangle \in \mathbb{C}^2$                                        | $ \psi\rangle,  \phi\rangle \in \mathbb{C}^2$                                        | by $\text{SWAP}( \psi\rangle \otimes  \phi\rangle) =  \phi\rangle \otimes  \psi\rangle$ for all $ \psi\rangle,  \phi\rangle \in \mathbb{C}^2$ . |
| rph8-g15q_F00060 | 5    | ■ 1.000 | $V_g \rho(x) V_g^\dagger = \rho(\alpha_g(x))$                                        | $V_g \rho(x) V_g^\dagger = \rho(\alpha_g(x))$                                        | be a map $\rho: \mathcal{X} \rightarrow \mathcal{S}(\mathcal{H})$ such that $V_g \rho(x) V_g^\dagger = \rho(\alpha_g(x))$ for                   |
| rph8-g15q_F00061 | 5    | ■ 1.000 | $x = (x_1, x_2) \in \mathcal{X}$                                                     | $x = (x_1, x_2) \in \mathcal{X}$                                                     | sends a point $x = (x_1, x_2) \in \mathcal{X}$ to                                                                                               |
| rph8-g15q_F00062 | 5    | ■ 1.000 | $Z$                                                                                  | $Z$                                                                                  | where $Z$ is the Pauli matrix $Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$ . From this unitary                                           |
| rph8-g15q_F00063 | 5    | ■ 1.000 | $Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$                                  | $Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$                                  | where $Z$ is the Pauli matrix $Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$ . From this unitary                                           |
| rph8-g15q_F00064 | 5    | ■ 1.000 | $0 < p < 1$                                                                          | $0 < p < 1$                                                                          | where $0 < p < 1$ . (The value of $p$ chosen to produce Fig. 3                                                                                  |
| rph8-g15q_F00065 | 5    | ■ 1.000 | $p$                                                                                  | $p$                                                                                  | where $0 < p < 1$ . (The value of $p$ chosen to produce Fig. 3                                                                                  |
| rph8-g15q_F00066 | 5    | ■ 1.000 | $ \pm, \pm\rangle :=  \pm\rangle \otimes  \pm\rangle$                                | $ \pm, \pm\rangle :=  \pm\rangle \otimes  \pm\rangle$                                | and $ \pm, \pm\rangle :=  \pm\rangle \otimes  \pm\rangle$ . Thus, the resulting state encoding                                                  |
| rph8-g15q_F00067 | 5    | ■ 0.994 | $x = (x_1, x_2)$                                                                     | $x = (x_1, x_2)$                                                                     | sends $x = (x_1, x_2)$ to                                                                                                                       |
| rph8-g15q_F00068 | 5    | ■ 1.000 | $\mathcal{L}: \mathbb{R}^2 \rightarrow \mathbb{M}_4$                                 | $\mathcal{L}: \mathbb{R}^2 \rightarrow \mathbb{M}_4$                                 | where $\mathcal{L}: \mathbb{R}^2 \rightarrow \mathbb{M}_4$ is a linear transformation sending each                                              |
| rph8-g15q_F00069 | 5    | ■ 1.000 | $\mathcal{L}(x) \in \mathbb{M}_4$                                                    | $\mathcal{L}(x) \in \mathbb{M}_4$                                                    | point $x \in \mathcal{X}$ to a Hermitian operator $\mathcal{L}(x) \in \mathbb{M}_4$ , where $\mathbb{M}_4$                                      |
| rph8-g15q_F00070 | 5    | ■ 1.000 | $\mathbb{M}_4$                                                                       | $\mathbb{M}_4$                                                                       | point $x \in \mathcal{X}$ to a Hermitian operator $\mathcal{L}(x) \in \mathbb{M}_4$ , where $\mathbb{M}_4$                                      |
| rph8-g15q_F00071 | 5    | ■ 0.968 | $-i\mathcal{L}(x) \in \mathfrak{su}(4)$                                              | $-i\mathcal{L}(x) \in \mathfrak{su}(4)$                                              | that $-i\mathcal{L}(x) \in \mathfrak{su}(4)$ , the Lie algebra of $4 \times 4$ special unitary                                                  |
| rph8-g15q_F00072 | 5    | ■ 0.968 | $4 \times 4$                                                                         | $4 \times 4$                                                                         | that $-i\mathcal{L}(x) \in \mathfrak{su}(4)$ , the Lie algebra of $4 \times 4$ special unitary                                                  |
| rph8-g15q_F00073 | 5    | ■ 1.000 | $\mathcal{L}$                                                                        | $\mathcal{L}$                                                                        | matrices. Such a linear transformation $\mathcal{L}$ is uniquely deter-                                                                         |
| rph8-g15q_F00074 | 5    | ■ 1.000 | $e_1 = (1, 0)$                                                                       | $e_1 = (1, 0)$                                                                       | where $e_1 = (1, 0)$ and $e_2 = (0, 1)$ . Therefore, the set of all                                                                             |
| rph8-g15q_F00075 | 5    | ■ 1.000 | $e_2 = (0, 1)$                                                                       | $e_2 = (0, 1)$                                                                       | where $e_1 = (1, 0)$ and $e_2 = (0, 1)$ . Therefore, the set of all                                                                             |
| rph8-g15q_F00076 | 5    | ■ 1.000 | $2(4^2 - 1) = 30$                                                                    | $2(4^2 - 1) = 30$                                                                    | a real vector space of dimension $2(4^2 - 1) = 30$ .                                                                                            |
| rph8-g15q_F00077 | 5    | ■ 1.000 | $\mathcal{L}(x)$                                                                     | $\mathcal{L}(x)$                                                                     | at the level of the generating elements $\mathcal{L}(x)$ because each $V_g$                                                                     |
| rph8-g15q_F00078 | 5    | ■ 1.000 | $V_g$                                                                                | $V_g$                                                                                | at the level of the generating elements $\mathcal{L}(x)$ because each $V_g$                                                                     |
| rph8-g15q_F00079 | 5    | ■ 1.000 | $g = (1, 0)$                                                                         | $g = (1, 0)$                                                                         | specific group elements. For $g = (1, 0)$ , equivariance implies                                                                                |
| rph8-g15q_F00080 | 5    | ■ 0.983 | $\text{SWAP}_{L_1} \text{SWAP} = L_2$                                                | $\text{SWAP}_{L_1} \text{SWAP} = L_2$                                                | $\text{SWAP } L_1 \text{ SWAP} = L_2$ (and also $\text{SWAP } L_2 \text{ SWAP} = L_1$ , but this                                                |
| rph8-g15q_F00081 | 5    | ■ 0.983 | $\text{SWAP}_{L_2} \text{SWAP} = L_1$                                                | $\text{SWAP}_{L_2} \text{SWAP} = L_1$                                                | $\text{SWAP } L_1 \text{ SWAP} = L_2$ (and also $\text{SWAP } L_2 \text{ SWAP} = L_1$ , but this                                                |
| rph8-g15q_F00082 | 5    | ■ 1.000 | $g = (0, 1)$                                                                         | $g = (0, 1)$                                                                         | is equivalent to the first equation). For $g = (0, 1)$ , equivari-                                                                              |
| rph8-g15q_F00083 | 5    | ■ 1.000 | $(X \otimes X)L_1(X \otimes X) = -L_1$                                               | $(X \otimes X)L_1(X \otimes X) = -L_1$                                               | ance implies $(X \otimes X)L_1(X \otimes X) = -L_1$ and $(X \otimes X)L_2(X \otimes$                                                            |
| rph8-g15q_F00084 | 5    | ■ 1.000 | $(X \otimes X)L_2(X \otimes$                                                         | $(X \otimes X)L_2(X \otimes$                                                         | ance implies $(X \otimes X)L_1(X \otimes X) = -L_1$ and $(X \otimes X)L_2(X \otimes$                                                            |
| rph8-g15q_F00085 | 5    | ■ 1.000 | $X) = -L_2$                                                                          | $X) = -L_2$                                                                          | $X) = -L_2$ . However, the second of these follows from the                                                                                     |



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| rph8-g15q_F00086 | 5    | 0.538 | $\left(L_{\{1\}}, L_{\{2\}}\right)$                                               | $(L_1, L_2)$                                                                | tian matrices $(L_1, L_2)$ to an 8-dimensional subspace of pairs                                                              |
| rph8-g15q_F00087 | 5    | 0.912 | $L_{\{1\}}$                                                                       | $L_1$                                                                       | $(L_1, \text{SWAP } L_1 \text{ SWAP})$ due to the constraint (18). A basis for this                                           |
| rph8-g15q_F00088 | 6    | 1.000 | $\tau_{\{\mathcal{X}\}}$                                                          | $\tau_{\mathcal{X}}$                                                        | is a set $\tau_{\mathcal{X}}$ whose elements are certain subsets of $\mathcal{X}$ , called                                    |
| rph8-g15q_F00089 | 6    | 0.999 | $\left(\mathcal{X}, \tau_{\{\mathcal{X}\}}\right)$                                | $(\mathcal{X}, \tau_{\mathcal{X}})$                                         | set $\mathcal{X}$ is equipped with a topology $\tau_{\mathcal{X}}$ , the pair $(\mathcal{X}, \tau_{\mathcal{X}})$ is          |
| rph8-g15q_F00090 | 6    | 1.000 | $\mathcal{A}$                                                                     | $\mathcal{A}$                                                               | mal smooth atlas $\mathcal{A}$ on $\mathcal{X}$ , and the triple $(\mathcal{X}, \tau_{\mathcal{X}}, \mathcal{A})$ is called a |
| rph8-g15q_F00091 | 6    | 1.000 | $\left(\mathcal{X}, \tau_{\{\mathcal{X}\}}, \mathcal{A}\right)$                   | $(\mathcal{X}, \tau_{\mathcal{X}}, \mathcal{A})$                            | mal smooth atlas $\mathcal{A}$ on $\mathcal{X}$ , and the triple $(\mathcal{X}, \tau_{\mathcal{X}}, \mathcal{A})$ is called a |
| rph8-g15q_F00092 | 6    | 0.755 | $\mathcal{H}=\mathbb{C}^d$                                                        | $\mathcal{H}=\mathbb{C}^d$                                                  | $\mathcal{H}=\mathbb{C}^d$ is the complex projective space $\mathbb{CP}^{d-1}$ , which also                                   |
| rph8-g15q_F00093 | 6    | 0.755 | $\mathbb{C}^{P^{d-1}}$                                                            | $\mathbb{CP}^{d-1}$                                                         | $\mathcal{H}=\mathbb{C}^d$ is the complex projective space $\mathbb{CP}^{d-1}$ , which also                                   |
| rph8-g15q_F00094 | 6    | 1.000 | $\mathcal{X}=\mathbb{R}^d$                                                        | $\mathcal{X}=\mathbb{R}^d$                                                  | for Hamiltonian evolution. In the case that $\mathcal{X}=\mathbb{R}^d$ is a Eu-                                               |
| rph8-g15q_F00095 | 6    | 1.000 | $x=\left(x_{\{1\}}, \ldots, x_{\{d\}}\right) \in$                                 | $x=\left(x_1, \ldots, x_d\right) \in$                                       | clidean feature space, each component of $x=\left(x_1, \ldots, x_d\right) \in$                                                |
| rph8-g15q_F00096 | 6    | 1.000 | $X_{\{k\}}$                                                                       | $X_k$                                                                       | where $X_k$ is the Pauli $X$ gate acting on the $k$ th qubit. The                                                             |
| rph8-g15q_F00097 | 6    | 1.000 | $k$                                                                               | $k$                                                                         | where $X_k$ is the Pauli $X$ gate acting on the $k$ th qubit. The                                                             |
| rph8-g15q_F00098 | 6    | 0.975 | $\rho: \mathbb{R}^d \rightarrow \mathcal{S}\left(\mathbb{C}^{2^d}\right)$         | $\rho: \mathbb{R}^d \rightarrow \mathcal{S}\left(\mathbb{C}^{2^d}\right)$   | associated quantum state encoding $\rho: \mathbb{R}^d \rightarrow \mathcal{S}\left(\mathbb{C}^{2^d}\right)$ is then           |
| rph8-g15q_F00099 | 6    | 1.000 | $Y$                                                                               | $Y$                                                                         | Note that the Pauli $X$ gate may be replaced with a Pauli $Y$ or                                                              |
| rph8-g15q_F00100 | 6    | 1.000 | $\vec{n} \cdot \vec{\sigma}$                                                      | $\vec{n} \cdot \vec{\sigma}$                                                | $\vec{n} \cdot \vec{\sigma}$ , where $\vec{n}$ is some unit vector in $\mathbb{R}^3$ and $\vec{\sigma}=(X, Y, Z)$             |
| rph8-g15q_F00101 | 6    | 1.000 | $\vec{n}$                                                                         | $\vec{n}$                                                                   | $\vec{n} \cdot \vec{\sigma}$ , where $\vec{n}$ is some unit vector in $\mathbb{R}^3$ and $\vec{\sigma}=(X, Y, Z)$             |
| rph8-g15q_F00102 | 6    | 1.000 | $\mathbb{R}^3$                                                                    | $\mathbb{R}^3$                                                              | $\vec{n} \cdot \vec{\sigma}$ , where $\vec{n}$ is some unit vector in $\mathbb{R}^3$ and $\vec{\sigma}=(X, Y, Z)$             |
| rph8-g15q_F00103 | 6    | 1.000 | $\vec{\sigma}=(X, Y, Z)$                                                          | $\vec{\sigma}=(X, Y, Z)$                                                    | $\vec{n} \cdot \vec{\sigma}$ , where $\vec{n}$ is some unit vector in $\mathbb{R}^3$ and $\vec{\sigma}=(X, Y, Z)$             |
| rph8-g15q_F00104 | 6    | 1.000 | $d$                                                                               | $d$                                                                         | tum encodings map a feature space of $d$ dimensions into $d$                                                                  |
| rph8-g15q_F00105 | 7    | 1.000 | $x \in \mathbb{R}^{2^d} \backslash\{0\}$                                          | $x \in \mathbb{R}^{2^d} \backslash\{0\}$                                    | [59]. By definition, it takes a vector $x \in \mathbb{R}^{2^d} \backslash\{0\}$ to a vector                                   |
| rph8-g15q_F00106 | 7    | 0.999 | $\chi(x)$                                                                         | $\chi(x)$                                                                   | $\chi(x)$ in $\mathbb{C}^{2^d}$ , the Hilbert space of $d$ qubits, and then normalizes                                        |
| rph8-g15q_F00107 | 7    | 0.999 | $\mathbb{C}^{2^d}$                                                                | $\mathbb{C}^{2^d}$                                                          | $\chi(x)$ in $\mathbb{C}^{2^d}$ , the Hilbert space of $d$ qubits, and then normalizes                                        |
| rph8-g15q_F00108 | 7    | 1.000 | $ \Phi(x)\rangle$                                                                 | $ \Phi(x)\rangle$                                                           | it to obtain a genuine quantum state $ \Phi(x)\rangle$ . To express this                                                      |
| rph8-g15q_F00109 | 7    | 1.000 | $x=\left(x_{\{0\}}, x_{\{1\}}, \ldots, x_{\{2^d-1\}}\right) \in \mathbb{R}^{2^d}$ | $x=\left(x_0, x_1, \ldots, x_{2^d-1}\right) \in \mathbb{R}^{2^d}$           | as a succinct formula, if $x=\left(x_0, x_1, \ldots, x_{2^d-1}\right) \in \mathbb{R}^{2^d}$ , then                            |
| rph8-g15q_F00110 | 7    | 0.998 | $n$                                                                               | $n$                                                                         | sequence $n$ of $d$ 0s and 1s. For example, if $d=2$ , then $n$                                                               |
| rph8-g15q_F00111 | 7    | 0.998 | $d \ 0 \ \mathrm{s}$                                                              | $d0 \ \mathrm{s}$                                                           | sequence $n$ of $d$ 0s and 1s. For example, if $d=2$ , then $n$                                                               |
| rph8-g15q_F00112 | 7    | 0.998 | $d=2$                                                                             | $d=2$                                                                       | sequence $n$ of $d$ 0s and 1s. For example, if $d=2$ , then $n$                                                               |
| rph8-g15q_F00113 | 7    | 0.999 | $n \in\{00,01,10,11\}$                                                            | $n \in\{00,01,10,11\}$                                                      | has four possibilities: $n \in\{00,01,10,11\}$ . Then, amplitude                                                              |
| rph8-g15q_F00114 | 7    | 1.000 | $\ \chi(x)\ $                                                                     | $\ \chi(x)\ $                                                               | where $\ \chi(x)\ $ denotes the Euclidean norm of $\chi(x)$ . Since the                                                       |
| rph8-g15q_F00115 | 7    | 1.000 | $S^{2^d-1}$                                                                       | $S^{2^d-1}$                                                                 | the data first onto the sphere $S^{2^d-1}$ inside $\mathbb{R}^{2^d}$ and then map                                             |
| rph8-g15q_F00116 | 7    | 1.000 | $\mathbb{R}^{2^d}$                                                                | $\mathbb{R}^{2^d}$                                                          | the data first onto the sphere $S^{2^d-1}$ inside $\mathbb{R}^{2^d}$ and then map                                             |
| rph8-g15q_F00117 | 7    | 0.999 | $\log _2\left(2^d\right)=d$                                                       | $\log _2\left(2^d\right)=d$                                                 | the form $\mathbb{R}^{2^d}$ is encoded into a Hilbert space of $\log _2\left(2^d\right)=d$                                    |
| rph8-g15q_F00118 | 7    | 1.000 | $S^{2^d}$                                                                         | $S^{2^d}$                                                                   | the original data and project onto the unit sphere $S^{2^d}$ (instead                                                         |
| rph8-g15q_F00119 | 7    | 1.000 | $\frac{\pi}{4}$                                                                   | $\frac{\pi}{4}$                                                             | hemisphere of $S^{2^d}$ within a polar angle of $\frac{\pi}{4}$ , while the points                                            |
| rph8-g15q_F00120 | 7    | 0.999 | $\frac{\pi}{2}$                                                                   | $\frac{\pi}{2}$                                                             | hemisphere of $S^{2^d}$ with a polar angle between $\frac{\pi}{4}$ and $\frac{\pi}{2}$ , thus                                 |
| rph8-g15q_F00121 | 7    | 0.952 | $x_{\{0\}}$                                                                       | $x_0$                                                                       | FIG. 5. Although data $x_0$ in the interval $[-1, 1]$ get mapped into                                                         |
| rph8-g15q_F00122 | 7    | 1.000 | $S^1$                                                                             | $S^1$                                                                       | the northern hemisphere of $S^1$ within polar angle $\frac{\pi}{4}$ , all other data                                          |
| rph8-g15q_F00123 | 7    | 1.000 | $[-6,6] \cap \mathbb{Z}$                                                          | $[-6,6] \cap \mathbb{Z}$                                                    | $[-6,6] \cap \mathbb{Z}$ .                                                                                                    |
| rph8-g15q_F00124 | 8    | 0.978 | $\mathbb{R}$                                                                      | $\mathbb{R}$                                                                | FIG. 6. Data $x_0$ in $\mathbb{R}$ get mapped into the probability 1-simplex.                                                 |
| rph8-g15q_F00125 | 8    | 0.706 | $\left(\mathcal{X}, d_{\mathcal{X}}\right)$                                       | $(\mathcal{X}, d_{\mathcal{X}})$                                            | <i>Definition 5.</i> A <i>metric space</i> is a pair $(\mathcal{X}, d_{\mathcal{X}})$ , where $\mathcal{X}$                   |
| rph8-g15q_F00126 | 8    | 1.000 | $d_{\mathcal{X}}: \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$          | $d_{\mathcal{X}}: \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$    | is a set and $d_{\mathcal{X}}: \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$ is a function satisfying the            |
| rph8-g15q_F00127 | 8    | 1.000 | $d_{\mathcal{X}}\left(x_1, x_2\right)=0$                                          | $d_{\mathcal{X}}\left(x_1, x_2\right)=0$                                    | (1) $d_{\mathcal{X}}\left(x_1, x_2\right)=0$ if and only if $x_1=x_2$ .                                                       |
| rph8-g15q_F00128 | 8    | 1.000 | $x_1=x_2$                                                                         | $x_1=x_2$                                                                   | (1) $d_{\mathcal{X}}\left(x_1, x_2\right)=0$ if and only if $x_1=x_2$ .                                                       |
| rph8-g15q_F00129 | 8    | 1.000 | $d_{\mathcal{X}}\left(x_1, x_2\right)=d_{\mathcal{X}}\left(x_2, x_1\right)$       | $d_{\mathcal{X}}\left(x_1, x_2\right)=d_{\mathcal{X}}\left(x_2, x_1\right)$ | (2) $d_{\mathcal{X}}\left(x_1, x_2\right)=d_{\mathcal{X}}\left(x_2, x_1\right)$ for all $x_1, x_2 \in \mathcal{X}$ .          |
| rph8-g15q_F00130 | 8    | 1.000 | $x_1, x_2 \in \mathcal{X}$                                                        | $x_1, x_2 \in \mathcal{X}$                                                  | (2) $d_{\mathcal{X}}\left(x_1, x_2\right)=d_{\mathcal{X}}\left(x_2, x_1\right)$ for all $x_1, x_2 \in \mathcal{X}$ .          |

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| rph8-g15q_F00131 | 8    | ■ 1.000 | $d_{\mathcal{X}}(\mathcal{X}) \leftarrow (x_1, x_2) \leq d_{\mathcal{X}}(x_1, x_3) + d_{\mathcal{X}}(x_2, x_3)$ | $d_{\mathcal{X}}(x_1, x_2) \leq d_{\mathcal{X}}(x_1, x_3) + d_{\mathcal{X}}(x_2, x_3)$ | (3) $d_{\mathcal{X}}(x_1, x_2) \leq d_{\mathcal{X}}(x_1, x_3) + d_{\mathcal{X}}(x_2, x_3)$ for all $x_1, x_2, x_3$                    |
| rph8-g15q_F00132 | 8    | ■ 1.000 | $x_1, x_2, x_3$                                                                                                 | $x_1, x_2, x_3$                                                                        | (3) $d_{\mathcal{X}}(x_1, x_2) \leq d_{\mathcal{X}}(x_1, x_3) + d_{\mathcal{X}}(x_2, x_3)$ for all $x_1, x_2, x_3$                    |
| rph8-g15q_F00133 | 8    | ■ 1.000 | $\in \mathcal{X}$                                                                                               | $\in \mathcal{X}$                                                                      | $\in \mathcal{X}$ (the <i>triangle inequality</i> ).                                                                                  |
| rph8-g15q_F00134 | 8    | ■ 1.000 | $d_{\mathcal{X}}$                                                                                               | $d_{\mathcal{X}}$                                                                      | Such a function $d_{\mathcal{X}}$ is called a <i>distance function</i> , or <i>metric</i> ,                                           |
| rph8-g15q_F00135 | 8    | ■ 1.000 | $X \subseteq \mathcal{X}$                                                                                       | $X \subseteq \mathcal{X}$                                                              | (1) If $X \subseteq \mathcal{X}$ is a finite subset of $\mathcal{X}$ , which is to be                                                 |
| rph8-g15q_F00136 | 8    | ■ 0.969 | $(X, d_X)$                                                                                                      | $(X, d_X)$                                                                             | distances $(X, d_X)$ to first construct a combinatorial object $K_{\bullet}(X, d_X)$ ,                                                |
| rph8-g15q_F00137 | 8    | ■ 0.969 | $K_{\bullet}(X, d_X)$                                                                                           | $K_{\bullet}(X, d_X)$                                                                  | distances $(X, d_X)$ to first construct a combinatorial object $K_{\bullet}(X, d_X)$ ,                                                |
| rph8-g15q_F00138 | 8    | ■ 1.000 | $H_{\bullet}(X, d_X)$                                                                                           | $H_{\bullet}(X, d_X)$                                                                  | which is then used to construct an algebraic object $H_{\bullet}(X, d_X)$ , which                                                     |
| rph8-g15q_F00139 | 8    | ■ 1.000 | $d_X$                                                                                                           | $d_X$                                                                                  | metric $d_{\mathcal{X}}$ on it, then $X$ acquires a metric $d_X$ by the formula                                                       |
| rph8-g15q_F00140 | 8    | ■ 1.000 | $d_X(x_1, x_2) = d_{\mathcal{X}}(x_1, x_2)$                                                                     | $d_X(x_1, x_2) = d_{\mathcal{X}}(x_1, x_2)$                                            | $d_X(x_1, x_2) = d_{\mathcal{X}}(x_1, x_2)$ for all $x_1, x_2 \in X$ .                                                                |
| rph8-g15q_F00141 | 8    | ■ 1.000 | $x_1, x_2 \in X$                                                                                                | $x_1, x_2 \in X$                                                                       | $d_X(x_1, x_2) = d_{\mathcal{X}}(x_1, x_2)$ for all $x_1, x_2 \in X$ .                                                                |
| rph8-g15q_F00142 | 8    | ■ 0.953 | $X, d_X$                                                                                                        | $X, d_X$                                                                               | (2) From $(X, d_X)$ , one can construct a combinatorial ob-                                                                           |
| rph8-g15q_F00143 | 8    | ■ 0.829 | $H_{\bullet}(X, d_X)$ [45,86,87]                                                                                | $H_{\bullet}(X, d_X)$ [45,86,87]                                                       | spaces and homologies $H_{\bullet}(X, d_X)$ [45,86,87].                                                                               |
| rph8-g15q_F00144 | 8    | ■ 0.920 | $\operatorname{dgm}_{\bullet}(X, d_X)$                                                                          | $\operatorname{dgm}_{\bullet}(X, d_X)$                                                 | numerical quantities, the persistence diagrams $\operatorname{dgm}_{\bullet}(X, d_X)$                                                 |
| rph8-g15q_F00145 | 8    | ■ 1.000 | $S$                                                                                                             | $S$                                                                                    | rather than the raw data. Briefly, if $S$ denotes a simplicial                                                                        |
| rph8-g15q_F00146 | 8    | ■ 1.000 | $\epsilon$                                                                                                      | $\epsilon$                                                                             | distance threshold $\epsilon$ , then each $k$ -simplex $s \in S$ is uniquely                                                          |
| rph8-g15q_F00147 | 8    | ■ 1.000 | $s \in S$                                                                                                       | $s \in S$                                                                              | distance threshold $\epsilon$ , then each $k$ -simplex $s \in S$ is uniquely                                                          |
| rph8-g15q_F00148 | 8    | ■ 1.000 | $k+1$                                                                                                           | $k+1$                                                                                  | determined by its $k+1$ vertices. Supposing that there is a                                                                           |
| rph8-g15q_F00149 | 8    | ■ 1.000 | $v_1, v_2, \dots, v_n$                                                                                          | $v_1, v_2, \dots, v_n$                                                                 | total ordering on the vertices, $v_1, v_2, \dots, v_n$ , such a simplex                                                               |
| rph8-g15q_F00150 | 8    | ■ 1.000 | $s$                                                                                                             | $s$                                                                                    | $s$ can therefore be expressed as $s = \{v_{i_1}, \dots, v_{i_{k+1}}\}$ , where                                                       |
| rph8-g15q_F00151 | 8    | ■ 1.000 | $s = \{v_{i_1}, \dots, v_{i_{k+1}}\}$                                                                           | $s = \{v_{i_1}, \dots, v_{i_{k+1}}\}$                                                  | $s$ can therefore be expressed as $s = \{v_{i_1}, \dots, v_{i_{k+1}}\}$ , where                                                       |
| rph8-g15q_F00152 | 8    | ■ 0.863 | $i_1 < \dots < i_{k+1}$                                                                                         | $i_1 < \dots < i_{k+1}$                                                                | $i_1 < \dots < i_{k+1}$ and the distinct vertices $v_{i_1}, \dots, v_{i_{k+1}}$ define                                                |
| rph8-g15q_F00153 | 8    | ■ 0.863 | $v_{i_1}, \dots, v_{i_{k+1}}$                                                                                   | $v_{i_1}, \dots, v_{i_{k+1}}$                                                          | $i_1 < \dots < i_{k+1}$ and the distinct vertices $v_{i_1}, \dots, v_{i_{k+1}}$ define                                                |
| rph8-g15q_F00154 | 8    | ■ 1.000 | $ s\rangle$                                                                                                     | $ s\rangle$                                                                            | the simplex $s$ . Therefore, the quantum state $ s\rangle$ associated with                                                            |
| rph8-g15q_F00155 | 8    | ■ 1.000 | $i_1, \dots, i_{k+1}$                                                                                           | $i_1, \dots, i_{k+1}$                                                                  | the simplex $s$ would be the $n$ -qubit state with the $i_1, \dots, i_{k+1}$                                                          |
| rph8-g15q_F00156 | 8    | ■ 1.000 | $ 1\rangle$                                                                                                     | $ 1\rangle$                                                                            | qubits in state $ 1\rangle$ and all other qubits in state $ 0\rangle$ (this is                                                        |
| rph8-g15q_F00157 | 8    | ■ 1.000 | $ 0\rangle$                                                                                                     | $ 0\rangle$                                                                            | qubits in state $ 1\rangle$ and all other qubits in state $ 0\rangle$ (this is                                                        |
| rph8-g15q_F00158 | 8    | ■ 1.000 | $n=7$                                                                                                           | $n=7$                                                                                  | tail in Appendix E). For example, for $n=7$ and $k=3$ , the                                                                           |
| rph8-g15q_F00159 | 8    | ■ 1.000 | $k=3$                                                                                                           | $k=3$                                                                                  | tail in Appendix E). For example, for $n=7$ and $k=3$ , the                                                                           |
| rph8-g15q_F00160 | 8    | ■ 1.000 | $s = \{v_2, v_4, v_5\}$                                                                                         | $s = \{v_2, v_4, v_5\}$                                                                | 2-simplex $s = \{v_2, v_4, v_5\}$ would be represented by the state                                                                   |
| rph8-g15q_F00161 | 8    | ■ 1.000 | $ s\rangle =  0101100\rangle$                                                                                   | $ s\rangle =  0101100\rangle$                                                          | $ s\rangle =  0101100\rangle$ .                                                                                                       |
| rph8-g15q_F00162 | 8    | ■ 1.000 | $f$                                                                                                             | $f$                                                                                    | <i>Definition 6.</i> A <i>distance nonincreasing</i> function $f$ :                                                                   |
| rph8-g15q_F00163 | 8    | ■ 1.000 | $(\mathcal{X}, d_{\mathcal{X}}) \rightarrow (\mathcal{Y}, d_{\mathcal{Y}})$                                     | $(\mathcal{X}, d_{\mathcal{X}}) \rightarrow (\mathcal{Y}, d_{\mathcal{Y}})$            | $(\mathcal{X}, d_{\mathcal{X}}) \rightarrow (\mathcal{Y}, d_{\mathcal{Y}})$ from one metric space $(\mathcal{X}, d_{\mathcal{X}})$ to |
| rph8-g15q_F00164 | 8    | ■ 1.000 | $(\mathcal{Y}, d_{\mathcal{Y}})$                                                                                | $(\mathcal{Y}, d_{\mathcal{Y}})$                                                       | another $(\mathcal{Y}, d_{\mathcal{Y}})$ is a function $f: \mathcal{X} \rightarrow \mathcal{Y}$ that satisfies                        |
| rph8-g15q_F00165 | 8    | ■ 1.000 | $f: \mathcal{X} \rightarrow \mathcal{Y}$                                                                        | $f: \mathcal{X} \rightarrow \mathcal{Y}$                                               | another $(\mathcal{Y}, d_{\mathcal{Y}})$ is a function $f: \mathcal{X} \rightarrow \mathcal{Y}$ that satisfies                        |
| rph8-g15q_F00166 | 8    | ■ 0.780 | $d_{\mathcal{Y}}(f(x_1), f(x_2)) \leq d_{\mathcal{X}}(x_1, x_2)$                                                | $d_{\mathcal{Y}}(f(x_1), f(x_2)) \leq d_{\mathcal{X}}(x_1, x_2)$                       | $d_{\mathcal{Y}}(f(x_1), f(x_2)) \leq d_{\mathcal{X}}(x_1, x_2)$ for all $x_1, x_2 \in \mathcal{X}$ . An <i>embed-</i>                |
| rph8-g15q_F00167 | 9    | ■ 1.000 | $d_{\mathcal{Y}}(f(x_1), f(x_2)) = d_{\mathcal{X}}(x_1, x_2)$                                                   | $d_{\mathcal{Y}}(f(x_1), f(x_2)) = d_{\mathcal{X}}(x_1, x_2)$                          | satisfies $d_{\mathcal{Y}}(f(x_1), f(x_2)) = d_{\mathcal{X}}(x_1, x_2)$ for all $x_1, x_2 \in \mathcal{X}$ ; i.e.,                    |
| rph8-g15q_F00168 | 9    | ■ 1.000 | $f: (X, d_X) \rightarrow (\mathcal{Y}, d_Y)$                                                                    | $f: (X, d_X) \rightarrow (\mathcal{Y}, d_Y)$                                           | metric $d_X$ from $d_{\mathcal{X}}$ . If $f: (X, d_X) \rightarrow (\mathcal{Y}, d_Y)$ is an embed-                                    |
| rph8-g15q_F00169 | 9    | ■ 1.000 | $Y := f(X)$                                                                                                     | $Y := f(X)$                                                                            | ding, let $Y := f(X)$ be the image of $X$ under $f$ equipped                                                                          |
| rph8-g15q_F00170 | 9    | ■ 1.000 | $d_Y$                                                                                                           | $d_Y$                                                                                  | with the induced metric $d_Y$ from $d_Y$ . Then, the filtered sim-                                                                    |
| rph8-g15q_F00171 | 9    | ■ 1.000 | $d_Y$                                                                                                           | $d_Y$                                                                                  | with the induced metric $d_Y$ from $d_Y$ . Then, the filtered sim-                                                                    |
| rph8-g15q_F00172 | 9    | ■ 1.000 | $K_{\bullet}(Y, d_Y)$                                                                                           | $K_{\bullet}(Y, d_Y)$                                                                  | plicial complexes $K_{\bullet}(X, d_X)$ and $K_{\bullet}(Y, d_Y)$ are isomorphic,                                                     |
| rph8-g15q_F00173 | 9    | ■ 1.000 | $H_{\bullet}(Y, d_Y)$                                                                                           | $H_{\bullet}(Y, d_Y)$                                                                  | the persistence vector spaces $H_{\bullet}(X, d_X)$ and $H_{\bullet}(Y, d_Y)$ are                                                     |
| rph8-g15q_F00174 | 9    | ■ 1.000 | $\operatorname{dgm}_{\bullet}(Y, d_Y)$                                                                          | $\operatorname{dgm}_{\bullet}(Y, d_Y)$                                                 | and $\operatorname{dgm}_{\bullet}(Y, d_Y)$ are all isomorphic.                                                                        |
| rph8-g15q_F00175 | 9    | ■ 0.988 | $\mathcal{X} \xrightarrow{f} \mathcal{Y}$                                                                       | $\mathcal{X} \xrightarrow{f} \mathcal{Y}$                                              | <i>Lemma 1.</i> Let $\mathcal{X}$ be a set, $(\mathcal{Y}, d_Y)$ a metric space, $\mathcal{X} \xrightarrow{f} \mathcal{Y}$            |



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| rph8-g15q_F00176 | 9    | ■ 1.000 | $d_{\{\mathrm{X}\}}: \mathrm{X} \times \mathrm{X} \rightarrow [0, \infty)$                                | $d_{\mathcal{X}}: \mathcal{X} \times \mathcal{X} \rightarrow [0, \infty)$           | a function, and $d_{\mathcal{X}}: \mathcal{X} \times \mathcal{X} \rightarrow [0, \infty)$ the function                                                                                                     |
| rph8-g15q_F00177 | 9    | ■ 1.000 | $\left(\mathrm{X}, d_{\{\mathrm{X}\}}\right) \xrightarrow{f} \left(\mathrm{Y}, d_{\{\mathrm{Y}\}}\right)$ | $(\mathcal{X}, d_{\mathcal{X}}) \xrightarrow{f} (\mathcal{Y}, d_{\mathcal{Y}})$     | only if $f$ is one to one. In such a case, $(\mathcal{X}, d_{\mathcal{X}}) \xrightarrow{f} (\mathcal{Y}, d_{\mathcal{Y}})$ is                                                                              |
| rph8-g15q_F00178 | 9    | ■ 1.000 | $L_{\{p\}}$                                                                                               | $L_p$                                                                               | metrics induced by the $L_p$ norm [91]) and (2) a family of                                                                                                                                                |
| rph8-g15q_F00179 | 10   | ■ 0.999 | $\left(\mathrm{X}, d_{\{\mathrm{X}_{\text{true}}\}}\right)$                                               | $(\mathcal{X}, d_{\mathcal{X}_{\text{true}}})$                                      | such that $(\mathcal{X}, d_{\mathcal{X}})$ is close to $(\mathcal{X}, d_{\mathcal{X}_{\text{true}}})$ , where $d_{\mathcal{X}_{\text{true}}}$ is some                                                      |
| rph8-g15q_F00180 | 10   | ■ 0.999 | $d_{\{\mathrm{X}_{\text{true}}\}}$                                                                        | $d_{\mathcal{X}_{\text{true}}}$                                                     | such that $(\mathcal{X}, d_{\mathcal{X}})$ is close to $(\mathcal{X}, d_{\mathcal{X}_{\text{true}}})$ , where $d_{\mathcal{X}_{\text{true}}}$ is some                                                      |
| rph8-g15q_F00181 | 10   | ■ 1.000 | $\sigma$                                                                                                  | $\sigma$                                                                            | <i>Definition 8.</i> Fix a Hilbert space $\mathcal{H}$ and let $\rho$ and $\sigma$ be two                                                                                                                  |
| rph8-g15q_F00182 | 10   | ■ 0.518 | $\ell_{\{1\}}$                                                                                            | $\ell_1$                                                                            | density matrices on $\mathcal{H}$ . The <i>trace/<math>\ell_1</math> distance</i> between $\rho$ and                                                                                                       |
| rph8-g15q_F00183 | 10   | ■ 1.000 | $\ell_{\{2\}}$                                                                                            | $\ell_2$                                                                            | The <i>Hilbert-Schmidt/Frobenius/<math>\ell_2</math> distance</i> is                                                                                                                                       |
| rph8-g15q_F00184 | 10   | ■ 1.000 | $\rho, \sigma \in \mathscr{S}(\mathrm{H})$                                                                | $\rho, \sigma \in \mathscr{S}(\mathcal{H})$                                         | denotes the <i>fidelity</i> between the states $\rho, \sigma \in \mathscr{S}(\mathcal{H})$ .                                                                                                               |
| rph8-g15q_F00185 | 10   | ■ 1.000 | $d_{\{\mathrm{H}\}}$                                                                                      | $d_{\mathcal{H}}$                                                                   | references therein. Let $d_{\mathcal{H}}$ be the notation for any one of                                                                                                                                   |
| rph8-g15q_F00186 | 10   | ■ 1.000 | $\Theta$                                                                                                  | $\Theta$                                                                            | One may additionally add on a parameter space $\Theta$ to define                                                                                                                                           |
| rph8-g15q_F00187 | 10   | ■ 1.000 | $\rho: \mathrm{X} \times \Theta \rightarrow \mathscr{S}(\mathrm{H})$                                      | $\rho: \mathcal{X} \times \Theta \rightarrow \mathscr{S}(\mathcal{H})$              | a function of the form $\rho: \mathcal{X} \times \Theta \rightarrow \mathscr{S}(\mathcal{H})$ , from which a                                                                                               |
| rph8-g15q_F00188 | 10   | ■ 1.000 | $\theta \in \Theta$                                                                                       | $\theta \in \Theta$                                                                 | specific parameter $\theta \in \Theta$ is chosen to optimize some machine                                                                                                                                  |
| rph8-g15q_F00189 | 10   | ■ 1.000 | $\mathrm{X}^{\{\prime\}}$                                                                                 | $\mathcal{X}'$                                                                      | for mathematical objects $\mathcal{X}'$ and $\mathscr{S}(\mathcal{H})'$ in a particular <i>cat-</i>                                                                                                        |
| rph8-g15q_F00190 | 10   | ■ 1.000 | $\mathscr{S}(\mathrm{H})^{\{\prime\}}$                                                                    | $\mathscr{S}(\mathcal{H})'$                                                         | for mathematical objects $\mathcal{X}'$ and $\mathscr{S}(\mathcal{H})'$ in a particular <i>cat-</i>                                                                                                        |
| rph8-g15q_F00191 | 10   | ■ 1.000 | $\rho^{\{\prime\}}: \mathrm{X}^{\{\prime\}} \rightarrow \mathscr{S}(\mathrm{H})^{\{\prime\}}$             | $\rho': \mathcal{X}' \rightarrow \mathscr{S}(\mathcal{H})'$                         | any structure, corresponds to a <i>morphism</i> $\rho': \mathcal{X}' \rightarrow \mathscr{S}(\mathcal{H})'$                                                                                                |
| rph8-g15q_F00192 | 10   | ■ 1.000 | $\rho^{\{\prime\}}$                                                                                       | $\rho'$                                                                             | in that category. This mapping $\rho'$ can be thought of as an                                                                                                                                             |
| rph8-g15q_F00193 | 11   | ■ 1.000 | $\mathrm{X}=[-2,2]$                                                                                       | $\mathcal{X}=[-2,2]$                                                                | FIG. 8. (a) Two classes of data on $\mathcal{X}=[-2,2]$ are presented. One class has orange data points labeled $\circ$ clustered near 0 and the other class                                               |
| rph8-g15q_F00194 | 11   | ■ 0.974 | $\rho: [-2,2] \rightarrow \mathscr{S}(\mathbb{C}^2)$                                                      | $\rho: [-2,2] \rightarrow \mathscr{S}(\mathbb{C}^2)$                                | has blue data points labeled $\times$ clustered near 2 and $-2$ . (b) The image of the quantum encoding $\rho: [-2,2] \rightarrow \mathscr{S}(\mathbb{C}^2)$ , with $\mathscr{S}(\mathbb{C}^2)$ visualized |
| rph8-g15q_F00195 | 11   | ■ 0.974 | $\mathscr{S}(\mathbb{C}^2)$                                                                               | $\mathscr{S}(\mathbb{C}^2)$                                                         | has blue data points labeled $\times$ clustered near 2 and $-2$ . (b) The image of the quantum encoding $\rho: [-2,2] \rightarrow \mathscr{S}(\mathbb{C}^2)$ , with $\mathscr{S}(\mathbb{C}^2)$ visualized |
| rph8-g15q_F00196 | 11   | ■ 0.614 | $\theta_1=0.31, \theta_2=1.48, \theta_3=0$                                                                | $\theta_1=0.31, \theta_2=1.48, \theta_3=0$                                          | as the Bloch ball and the encoding $\rho$ defined in Eq. (D10) using the optimized parameters $\theta_1=0.31, \theta_2=1.48, \theta_3=0$ , and $\theta_4=0$ . (c) A                                        |
| rph8-g15q_F00197 | 11   | ■ 0.614 | $\theta_4=0$                                                                                              | $\theta_4=0$                                                                        | as the Bloch ball and the encoding $\rho$ defined in Eq. (D10) using the optimized parameters $\theta_1=0.31, \theta_2=1.48, \theta_3=0$ , and $\theta_4=0$ . (c) A                                        |
| rph8-g15q_F00198 | 11   | ■ 0.993 | $\rho: \mathrm{X} \rightarrow \mathscr{S}(\mathbb{C}^2)$                                                  | $\rho: \mathcal{X} \rightarrow \mathscr{S}(\mathbb{C}^2)$                           | visualization of the pullback metric from the quantum encoding $\rho: \mathcal{X} \rightarrow \mathscr{S}(\mathbb{C}^2)$ back onto the domain $\mathcal{X}$ . The metric has been uniformly                |
| rph8-g15q_F00199 | 11   | ■ 0.679 | $f:$                                                                                                      | $f:$                                                                                | that for any two sets $\mathcal{X}$ and $\mathcal{Y}$ , there is a set of functions $f:$                                                                                                                   |
| rph8-g15q_F00200 | 11   | ■ 1.000 | $\mathrm{X} \rightarrow \mathrm{Y}$                                                                       | $\mathcal{X} \rightarrow \mathcal{Y}$                                               | $\mathcal{X} \rightarrow \mathcal{Y}$ from $\mathcal{X}$ to $\mathcal{Y}$ . Functions, though quite minimalistic,                                                                                          |
| rph8-g15q_F00201 | 11   | ■ 1.000 | $g: \mathrm{Y} \rightarrow \mathrm{Z}$                                                                    | $g: \mathcal{Y} \rightarrow \mathcal{Z}$                                            | Namely, for any two functions $f: \mathcal{X} \rightarrow \mathcal{Y}$ and $g: \mathcal{Y} \rightarrow \mathcal{Z}$ ,                                                                                      |
| rph8-g15q_F00202 | 11   | ■ 1.000 | $g \circ f: \mathrm{X} \rightarrow \mathrm{Z}$                                                            | $g \circ f: \mathcal{X} \rightarrow \mathcal{Z}$                                    | their composite is also a function $g \circ f: \mathcal{X} \rightarrow \mathcal{Z}$ . Moreover,                                                                                                            |
| rph8-g15q_F00203 | 11   | ■ 1.000 | $f: \mathrm{X} \rightarrow \mathrm{Y}, g: \mathrm{Y} \rightarrow \mathrm{Z}$                              | $f: \mathcal{X} \rightarrow \mathcal{Y}, g: \mathcal{Y} \rightarrow \mathcal{Z}$    | functions $f: \mathcal{X} \rightarrow \mathcal{Y}, g: \mathcal{Y} \rightarrow \mathcal{Z}$ , and $h: \mathcal{Z} \rightarrow \mathcal{W}$ , the com-                                                       |
| rph8-g15q_F00204 | 11   | ■ 1.000 | $h: \mathrm{Z} \rightarrow \mathrm{W}$                                                                    | $h: \mathcal{Z} \rightarrow \mathcal{W}$                                            | functions $f: \mathcal{X} \rightarrow \mathcal{Y}, g: \mathcal{Y} \rightarrow \mathcal{Z}$ , and $h: \mathcal{Z} \rightarrow \mathcal{W}$ , the com-                                                       |
| rph8-g15q_F00205 | 11   | ■ 1.000 | $h \circ (g \circ f)$                                                                                     | $h \circ (g \circ f)$                                                               | posites $h \circ (g \circ f)$ and $(h \circ g) \circ f$ are equal. Finally, for each                                                                                                                       |
| rph8-g15q_F00206 | 11   | ■ 1.000 | $(h \circ g) \circ f$                                                                                     | $(h \circ g) \circ f$                                                               | posites $h \circ (g \circ f)$ and $(h \circ g) \circ f$ are equal. Finally, for each                                                                                                                       |
| rph8-g15q_F00207 | 11   | ■ 1.000 | $\operatorname{id}_{\{\mathrm{X}\}}: \mathrm{X} \rightarrow \mathrm{X}$                                   | $\operatorname{id}_{\mathcal{X}}: \mathcal{X} \rightarrow \mathcal{X}$              | set $\mathcal{X}$ , there is a particular function $\operatorname{id}_{\mathcal{X}}: \mathcal{X} \rightarrow \mathcal{X}$ defined by                                                                       |
| rph8-g15q_F00208 | 11   | ■ 0.996 | $\operatorname{id}_{\{\mathrm{X}\}}(x)=x$                                                                 | $\operatorname{id}_{\mathcal{X}}(x)=x$                                              | $\operatorname{id}_{\mathcal{X}}(x)=x$ for all $x \in \mathcal{X}$ that acts as a unit/identity for this                                                                                                   |
| rph8-g15q_F00209 | 11   | ■ 0.612 | $\operatorname{id}_{\{\mathrm{Y}\}} \circ f=f \circ \operatorname{id}_{\{\mathrm{X}\}}=f$                 | $\operatorname{id}_{\mathcal{Y}} \circ f=f \circ \operatorname{id}_{\mathcal{X}}=f$ | composition since $\operatorname{id}_{\mathcal{Y}} \circ f=f \circ \operatorname{id}_{\mathcal{X}}=f$ for all functions                                                                                    |
| rph8-g15q_F00210 | 11   | ■ 1.000 | $\mathbf{C}$                                                                                              | $\mathbf{C}$                                                                        | <i>Definition 10.</i> A category $\mathbf{C}$ consists of                                                                                                                                                  |
| rph8-g15q_F00211 | 11   | ■ 0.996 | $\mathrm{X}, \mathrm{Y}, \ldots$                                                                          | $\mathcal{X}, \mathcal{Y}, \ldots$                                                  | (1) a collection of <i>objects</i> , denoted as $\mathcal{X}, \mathcal{Y}, \ldots$                                                                                                                         |
| rph8-g15q_F00212 | 11   | ■ 1.000 | $\mathrm{X}, \mathrm{Y}$                                                                                  | $\mathcal{X}, \mathcal{Y}$                                                          | (2) for every pair of objects $\mathcal{X}, \mathcal{Y}$ , a set of <i>morphisms</i> from                                                                                                                  |
| rph8-g15q_F00213 | 11   | ■ 0.957 | $f: \mathrm{X} \rightarrow \mathrm{Y}, g: \mathrm{Y} \rightarrow \mathrm{Z}$                              | $f: \mathcal{X} \rightarrow \mathcal{Y}, g: \mathcal{Y} \rightarrow \mathcal{Z}$ ,  | composable morphisms $f: \mathcal{X} \rightarrow \mathcal{Y}, g: \mathcal{Y} \rightarrow \mathcal{Z}$ , and                                                                                                |
| rph8-g15q_F00214 | 12   | ■ 0.737 | $f \circ \operatorname{id}_{\{\mathrm{X}\}}=$                                                             | $f \circ \operatorname{id}_{\mathcal{X}}=$                                          | $\mathcal{X}$ there exists a morphism $\operatorname{id}_{\mathcal{X}}: \mathcal{X} \rightarrow \mathcal{X}$ such that $f \circ \operatorname{id}_{\mathcal{X}}=$                                          |
| rph8-g15q_F00215 | 12   | ■ 0.994 | $\operatorname{id}_{\{\mathrm{Y}\}} \circ f=f$                                                            | $\operatorname{id}_{\mathcal{Y}} \circ f=f$                                         | $\operatorname{id}_{\mathcal{Y}} \circ f=f$ for all morphisms $f: \mathcal{X} \rightarrow \mathcal{Y}$ .                                                                                                   |
| rph8-g15q_F00216 | 12   | ■ 0.932 | $\operatorname{id}_{\{\mathrm{X}\}}$                                                                      | $\operatorname{id}_{\mathcal{X}}$                                                   | (3) For each $G$ -set $(\mathcal{X}, \alpha)$ , the identity function $\operatorname{id}_{\mathcal{X}}$ on $\mathcal{X}$                                                                                   |
| rph8-g15q_F00217 | 12   | ■ 0.985 | $\operatorname{emb}$                                                                                      | $\operatorname{emb}$                                                                | category $\mathbf{Met}^{\operatorname{emb}}$ whose objects are metric spaces, where a                                                                                                                      |
| rph8-g15q_F00218 | 12   | ■ 0.958 | $\operatorname{nif}$                                                                                      | $\operatorname{nif}$                                                                | <i>functions</i> ). There is a category $\mathbf{Met}^{\operatorname{nif}}$ whose objects are met-                                                                                                         |

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| rph8-g15q_F00219 | 12   | 0.738 | $\mathbf{Met}^{\mathrm{nif}}$                                                                 | $\mathbf{Met}^{\mathrm{nif}}$                                                                 | ric spaces $(\mathcal{X}, d_{\mathcal{X}})$ . A morphism in $\mathbf{Met}^{\mathrm{nif}}$ between metric                            |
| rph8-g15q_F00220 | 12   | 0.982 | $\{\}^{\mathrm{inf}}$                                                                         | $\mathrm{inf}$                                                                                | <i>subcategory</i> of $\mathbf{Met}^{\mathrm{inf}}$ , meaning that all objects and mor-                                             |
| rph8-g15q_F00221 | 12   | 0.876 | $\{\}^{\mathrm{nif}}$                                                                         | $\mathrm{nif}$                                                                                | phisms of $\mathbf{Met}^{\mathrm{emb}}$ are also objects and morphisms of $\mathbf{Met}^{\mathrm{nif}}$ ,                           |
| rph8-g15q_F00222 | 12   | 1.000 | $\rho: (\mathcal{X}, \alpha) \rightarrow (\mathcal{S}(\mathcal{H}), \mathrm{Ad}_V)$           | $\rho: (\mathcal{X}, \alpha) \rightarrow (\mathcal{S}(\mathcal{H}), \mathrm{Ad}_V)$           | a $G$ -equivariant function $\rho: (\mathcal{X}, \alpha) \rightarrow (\mathcal{S}(\mathcal{H}), \mathrm{Ad}_V)$ which,              |
| rph8-g15q_F00223 | 12   | 0.999 | $F: G$                                                                                        | $F: G$                                                                                        | denote this assignment by $F: G\text{-}\mathbf{Set} \rightarrow \mathbf{Set}$ , so that for an                                      |
| rph8-g15q_F00224 | 13   | 1.000 | $F(\mathcal{X}, \alpha) = \mathcal{X}$                                                        | $F(\mathcal{X}, \alpha) = \mathcal{X}$                                                        | arbitrary $G$ -set $(\mathcal{X}, \alpha)$ , we have $F(\mathcal{X}, \alpha) = \mathcal{X}$ and for any $G$ -                       |
| rph8-g15q_F00225 | 13   | 1.000 | $F(f) = f$                                                                                    | $F(f) = f$                                                                                    | equivariant function $f$ , we have $F(f) = f$ . Then, $F$ behaves                                                                   |
| rph8-g15q_F00226 | 13   | 1.000 | $F$                                                                                           | $F$                                                                                           | equivariant function $f$ , we have $F(f) = f$ . Then, $F$ behaves                                                                   |
| rph8-g15q_F00227 | 13   | 1.000 | $g$                                                                                           | $g$                                                                                           | and $g$ , we have, somewhat trivially,                                                                                              |
| rph8-g15q_F00228 | 13   | 1.000 | $x \mapsto x$                                                                                 | $x \mapsto x$                                                                                 | finied by $x \mapsto x$ for all $x \in \mathcal{X}$ . So, forgetting the group action                                               |
| rph8-g15q_F00229 | 13   | 1.000 | $F: \mathbf{C} \rightarrow \mathbf{D}$                                                        | $F: \mathbf{C} \rightarrow \mathbf{D}$                                                        | <i>Definition 11.</i> A <i>functor</i> $F: \mathbf{C} \rightarrow \mathbf{D}$ between categories                                    |
| rph8-g15q_F00230 | 13   | 0.582 | $F(\mathcal{X})$                                                                              | $F(\mathcal{X})$                                                                              | (1) an object $F(\mathcal{X})$ in $\mathbf{D}$ for every object $\mathcal{X}$ in $\mathbf{C}$                                       |
| rph8-g15q_F00231 | 13   | 0.582 | $\mathbf{D}$                                                                                  | $\mathbf{D}$                                                                                  | (1) an object $F(\mathcal{X})$ in $\mathbf{D}$ for every object $\mathcal{X}$ in $\mathbf{C}$                                       |
| rph8-g15q_F00232 | 13   | 0.988 | $F(f): F(\mathcal{X}) \rightarrow F(\mathcal{Y})$                                             | $F(f): F(\mathcal{X}) \rightarrow F(\mathcal{Y})$                                             | (2) a morphism $F(f): F(\mathcal{X}) \rightarrow F(\mathcal{Y})$ in $\mathbf{D}$ for every                                          |
| rph8-g15q_F00233 | 13   | 1.000 | $F(g \circ f) =$                                                                              | $F(g \circ f) =$                                                                              | (1) Composition is preserved; that is, $F(g \circ f) =$                                                                             |
| rph8-g15q_F00234 | 13   | 1.000 | $F(g) \circ F(f)$                                                                             | $F(g) \circ F(f)$                                                                             | $F(g) \circ F(f)$ for all morphisms $f: \mathcal{X} \rightarrow \mathcal{Y}$ and $g: \mathcal{Y} \rightarrow \mathcal{Z}$           |
| rph8-g15q_F00235 | 13   | 0.452 | $F(\mathrm{id}_{\mathcal{X}}) = \mathrm{id}_{F(\mathcal{X})}$                                 | $F(\mathrm{id}_{\mathcal{X}}) = \mathrm{id}_{F(\mathcal{X})}$                                 | (2) Identities are preserved; that is, $F(\mathrm{id}_{\mathcal{X}}) = \mathrm{id}_{F(\mathcal{X})}$ for                            |
| rph8-g15q_F00236 | 13   | 1.000 | $\mathbf{C}(\mathcal{X}, \mathcal{Y})$                                                        | $\mathbf{C}(\mathcal{X}, \mathcal{Y})$                                                        | properties that functors may satisfy. Let $\mathbf{C}(\mathcal{X}, \mathcal{Y})$ denote the                                         |
| rph8-g15q_F00237 | 13   | 1.000 | $\mathbf{C}(\mathcal{X}, \mathcal{Y}) \rightarrow \mathbf{D}(F(\mathcal{X}), F(\mathcal{Y}))$ | $\mathbf{C}(\mathcal{X}, \mathcal{Y}) \rightarrow \mathbf{D}(F(\mathcal{X}), F(\mathcal{Y}))$ | $\mathbf{C}(\mathcal{X}, \mathcal{Y}) \rightarrow \mathbf{D}(F(\mathcal{X}), F(\mathcal{Y}))$ defined by $f \mapsto F(f)$ is injec- |
| rph8-g15q_F00238 | 13   | 1.000 | $f \mapsto F(f)$                                                                              | $f \mapsto F(f)$                                                                              | $\mathbf{C}(\mathcal{X}, \mathcal{Y}) \rightarrow \mathbf{D}(F(\mathcal{X}), F(\mathcal{Y}))$ defined by $f \mapsto F(f)$ is injec- |
| rph8-g15q_F00239 | 13   | 0.808 | $\mathbf{C} \rightarrow \mathbf{D}$                                                           | $\mathbf{C} \rightarrow \mathbf{D}$                                                           | Loosely speaking, a functor $\mathbf{C} \rightarrow \mathbf{D}$ is called <i>forgetful</i> if it                                    |
| rph8-g15q_F00240 | 14   | 1.000 | $F(\mathcal{X}') = \mathcal{X}$                                                               | $F(\mathcal{X}') = \mathcal{X}$                                                               | $F(\mathcal{X}') = \mathcal{X}$ and $F(\mathcal{S}(\mathcal{H})) = \mathcal{S}(\mathcal{H})$ are in the image of a                  |
| rph8-g15q_F00241 | 14   | 1.000 | $F(\mathcal{S}(\mathcal{H})) = \mathcal{S}(\mathcal{H})$                                      | $F(\mathcal{S}(\mathcal{H})) = \mathcal{S}(\mathcal{H})$                                      | $F(\mathcal{X}') = \mathcal{X}$ and $F(\mathcal{S}(\mathcal{H})) = \mathcal{S}(\mathcal{H})$ are in the image of a                  |
| rph8-g15q_F00242 | 14   | 0.880 | $F: \mathbf{C} \rightarrow$                                                                   | $F: \mathbf{C} \rightarrow$                                                                   | faithful forgetful functor $F: \mathbf{C} \rightarrow \mathbf{Set}$ .                                                               |
| rph8-g15q_F00243 | 14   | 1.000 | $F(\rho') = \rho$                                                                             | $F(\rho') = \rho$                                                                             | $\mathbf{C}$ such that $F(\rho') = \rho$ . The morphism $\rho'$ is the desired,                                                     |
| rph8-g15q_F00244 | 14   | 0.999 | $\mathcal{X}' := (\mathcal{X}, \alpha)$                                                       | $\mathcal{X}' := (\mathcal{X}, \alpha)$                                                       | <i>Example 11.</i> Let $\mathcal{X}' := (\mathcal{X}, \alpha)$ be a data domain that is                                             |
| rph8-g15q_F00245 | 14   | 1.000 | $G \ni g \mapsto V_g \in \mathcal{U}(\mathcal{H})$                                            | $G \ni g \mapsto V_g \in \mathcal{U}(\mathcal{H})$                                            | action. Also, if $G \ni g \mapsto V_g \in \mathcal{U}(\mathcal{H})$ is a unitary representa-                                        |
| rph8-g15q_F00246 | 14   | 1.000 | $G \ni g \mapsto \mathrm{Ad}_{V_g}$                                                           | $G \ni g \mapsto \mathrm{Ad}_{V_g}$                                                           | action of $G$ on $\mathcal{S}(\mathcal{H})$ via $G \ni g \mapsto \mathrm{Ad}_{V_g}$ , then $\mathcal{S}(\mathcal{H})' :=$           |
| rph8-g15q_F00247 | 14   | 1.000 | $\mathcal{S}(\mathcal{H})' :=$                                                                | $\mathcal{S}(\mathcal{H})' :=$                                                                | action of $G$ on $\mathcal{S}(\mathcal{H})$ via $G \ni g \mapsto \mathrm{Ad}_{V_g}$ , then $\mathcal{S}(\mathcal{H})' :=$           |
| rph8-g15q_F00248 | 14   | 0.902 | $\rho': \mathcal{X}' \rightarrow$                                                             | $\rho': \mathcal{X}' \rightarrow$                                                             | then the quantum encoding “lifts” to a morphism $\rho': \mathcal{X}' \rightarrow$                                                   |
| rph8-g15q_F00249 | 14   | 1.000 | $U$                                                                                           | $U$                                                                                           | the structure preserved under quantum unitary encodings $U:$                                                                        |
| rph8-g15q_F00250 | 14   | 1.000 | $\mathbb{R}^d \rightarrow \mathcal{U}(\mathcal{H})$                                           | $\mathbb{R}^d \rightarrow \mathcal{U}(\mathcal{H})$                                           | $\mathbb{R}^d \rightarrow \mathcal{U}(\mathcal{H})$ of the form                                                                     |
| rph8-g15q_F00251 | 14   | 1.000 | $\mathcal{L}: \mathbb{R}^d \rightarrow \mathcal{B}(\mathcal{H})$                              | $\mathcal{L}: \mathbb{R}^d \rightarrow \mathcal{B}(\mathcal{H})$                              | where $\mathcal{L}: \mathbb{R}^d \rightarrow \mathcal{B}(\mathcal{H})$ is a linear transformation sending                           |
| rph8-g15q_F00252 | 14   | 1.000 | $x \in \mathbb{R}^d$                                                                          | $x \in \mathbb{R}^d$                                                                          | each point $x \in \mathbb{R}^d$ to a Hermitian operator $\mathcal{L}(x)$ acting on                                                  |
| rph8-g15q_F00253 | 14   | 1.000 | $\mathcal{B}(\mathcal{H})$                                                                    | $\mathcal{B}(\mathcal{H})$                                                                    | a Hilbert space $\mathcal{H}$ [here, $\mathcal{B}(\mathcal{H})$ denotes the set of bounded                                          |
| rph8-g15q_F00254 | 14   | 1.000 | $U: \mathbb{R}^d \rightarrow \mathcal{U}(\mathcal{H})$                                        | $U: \mathbb{R}^d \rightarrow \mathcal{U}(\mathcal{H})$                                        | $U: \mathbb{R}^d \rightarrow \mathcal{U}(\mathcal{H})$ that are morphisms in <b>LieOPS</b> (i.e., that                              |
| rph8-g15q_F00255 | 15   | 1.000 | $\rho: X \rightarrow \mathcal{S}(\mathcal{H})$                                                | $\rho: X \rightarrow \mathcal{S}(\mathcal{H})$                                                | ing $\rho: X \rightarrow \mathcal{S}(\mathcal{H})$ from the space of data points $X$ to the                                         |
| rph8-g15q_F00256 | 15   | 1.000 | $\rho_{\mathrm{opt}}$                                                                         | $\rho_{\mathrm{opt}}$                                                                         | An optimal solution $\rho_{\mathrm{opt}}$ in the context of quantum encodings                                                       |
| rph8-g15q_F00257 | 16   | 0.995 | $\mathcal{L}: \mathcal{X} \rightarrow \mathcal{B}(\mathcal{H})$                               | $\mathcal{L}: \mathcal{X} \rightarrow \mathcal{B}(\mathcal{H})$                               | where $\mathcal{L}: \mathcal{X} \rightarrow \mathcal{B}(\mathcal{H})$ is a linear transformation sending each                       |
| rph8-g15q_F00258 | 16   | 1.000 | $\rho(x) = U(x)  \psi_0\rangle \langle \psi_0  U(x)^\dagger$                                  | $\rho(x) = U(x)  \psi_0\rangle \langle \psi_0  U(x)^\dagger$                                  | $ \psi_0\rangle \in \mathcal{H}$ via the formula $\rho(x) = U(x)  \psi_0\rangle \langle \psi_0  U(x)^\dagger$ . Exam-               |
| rph8-g15q_F00259 | 16   | 1.000 | $\gamma: \mathbb{R} \rightarrow G$                                                            | $\gamma: \mathbb{R} \rightarrow G$                                                            | morphism $\gamma: \mathbb{R} \rightarrow G$ , where $\mathbb{R}$ is viewed as a group under                                         |
| rph8-g15q_F00260 | 16   | 1.000 | $\gamma$                                                                                      | $\gamma$                                                                                      | (1) $\gamma$ is a continuous function,                                                                                              |
| rph8-g15q_F00261 | 16   | 1.000 | $\gamma(0) = 1$                                                                               | $\gamma(0) = 1$                                                                               | (2) $\gamma(0) = 1$ , and                                                                                                           |
| rph8-g15q_F00262 | 16   | 1.000 | $\gamma(s+t) = \gamma(s)\gamma(t)$                                                            | $\gamma(s+t) = \gamma(s)\gamma(t)$                                                            | (3) $\gamma(s+t) = \gamma(s)\gamma(t)$ for all $s, t \in \mathbb{R}$ .                                                              |
| rph8-g15q_F00263 | 16   | 1.000 | $s, t \in \mathbb{R}$                                                                         | $s, t \in \mathbb{R}$                                                                         | (3) $\gamma(s+t) = \gamma(s)\gamma(t)$ for all $s, t \in \mathbb{R}$ .                                                              |
| rph8-g15q_F00264 | 16   | 1.000 | $M \in \mathfrak{g}$                                                                          | $M \in \mathfrak{g}$                                                                          | Lie group $G$ is generated by a unique element $M \in \mathfrak{g}$ in the                                                          |



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| rph8-g15q_F00265 | 16   | ■ 0.999 | $\mathfrak{g}$                                                                    | $\mathfrak{g}$                                                                     | Lie algebra $\mathfrak{g}$ by the exponential map $\exp : \mathfrak{g} \rightarrow G$ . Namely,                             |
| rph8-g15q_F00266 | 16   | ■ 0.999 | $\exp : \mathfrak{g} \rightarrow G$                                               | $\exp : \mathfrak{g} \rightarrow G$                                                | Lie algebra $\mathfrak{g}$ by the exponential map $\exp : \mathfrak{g} \rightarrow G$ . Namely,                             |
| rph8-g15q_F00267 | 16   | ■ 1.000 | $t \in \mathbb{R}$                                                                | $t \in \mathbb{R}$                                                                 | for all $t \in \mathbb{R}$ . In particular, this implies $\gamma$ is not only continu-                                      |
| rph8-g15q_F00268 | 16   | ■ 1.000 | $t=0$                                                                             | $t = 0$                                                                            | $\gamma$ at $t = 0$ (cf. Sec. 1.3 in Ref. [71]), one obtains $M = \frac{d}{dt}\gamma _{t=0}$ .                              |
| rph8-g15q_F00269 | 16   | ■ 1.000 | $M=\left.\frac{d}{dt}\gamma\right _{t=0}$                                         | $M = \left.\frac{d}{dt}\gamma\right _{t=0}$                                        | $\gamma$ at $t = 0$ (cf. Sec. 1.3 in Ref. [71]), one obtains $M = \frac{d}{dt}\gamma _{t=0}$ .                              |
| rph8-g15q_F00270 | 16   | ■ 1.000 | $\mathcal{X}=\mathbb{R}$                                                          | $\mathcal{X} = \mathbb{R}$                                                         | domain $\mathcal{X}$ of the form $\mathcal{X} = \mathbb{R}$ , since in this case $x \mapsto e^{-ixL}$ ,                     |
| rph8-g15q_F00271 | 16   | ■ 1.000 | $x \mapsto e^{-i x L}$                                                            | $x \mapsto e^{-ixL}$                                                               | domain $\mathcal{X}$ of the form $\mathcal{X} = \mathbb{R}$ , since in this case $x \mapsto e^{-ixL}$ ,                     |
| rph8-g15q_F00272 | 16   | ■ 0.999 | $L \in \mathfrak{su}(2^n)$                                                        | $L \in \mathfrak{su}(2^n)$                                                         | with $L \in \mathfrak{su}(2^n)$ for a quantum unitary encoding mapping                                                      |
| rph8-g15q_F00273 | 16   | ■ 0.995 | $\exp : \mathbb{R}^d \rightarrow \mathbb{R}^d$                                    | $\exp : \mathbb{R}^d \rightarrow \mathbb{R}^d$                                     | noting that the exponential map $\exp : \mathbb{R}^d \rightarrow \mathbb{R}^d$ from the Lie                                 |
| rph8-g15q_F00274 | 16   | ■ 1.000 | $\exp (x)=x$                                                                      | $\exp(x) = x$                                                                      | $\mathbb{R}^d$ is the identity map, $\exp(x) = x$ for all $x \in \mathbb{R}^d$ .                                            |
| rph8-g15q_F00275 | 16   | ■ 1.000 | $[V, W]:=V W-W V$                                                                 | $[V, W] := VW - WV$                                                                | $[V, W] := VW - WV$ denotes the commutator.)                                                                                |
| rph8-g15q_F00276 | 16   | ■ 1.000 | $H$                                                                               | $H$                                                                                | <i>Theorem A1.</i> Let $G$ be any connected Lie group, let $H$ be                                                           |
| rph8-g15q_F00277 | 16   | ■ 1.000 | $f: G \rightarrow H$                                                              | $f : G \rightarrow H$                                                              | any Lie group, and let $f : G \rightarrow H$ be a Lie group homomor-                                                        |
| rph8-g15q_F00278 | 16   | ■ 0.979 | $\mathcal{M}: \mathfrak{g} \rightarrow \mathfrak{h}$                              | $\mathcal{M} : \mathfrak{g} \rightarrow \mathfrak{h}$                              | phism. Then, there exists a unique real-linear map $\mathcal{M} : \mathfrak{g} \rightarrow \mathfrak{h}$                    |
| rph8-g15q_F00279 | 16   | ■ 1.000 | $W \in \mathfrak{g}$                                                              | $W \in \mathfrak{g}$                                                               | for all $W \in \mathfrak{g}$ . Moreover, $\mathcal{M}$ is a Lie algebra homomorphism,                                       |
| rph8-g15q_F00280 | 16   | ■ 1.000 | $\mathcal{M}$                                                                     | $\mathcal{M}$                                                                      | for all $W \in \mathfrak{g}$ . Moreover, $\mathcal{M}$ is a Lie algebra homomorphism,                                       |
| rph8-g15q_F00281 | 16   | ■ 1.000 | $W_1, W_2 \in \mathfrak{g}$                                                       | $W_1, W_2 \in \mathfrak{g}$                                                        | for all $W_1, W_2 \in \mathfrak{g}$ .                                                                                       |
| rph8-g15q_F00282 | 17   | ■ 1.000 | $U: \mathcal{X} \rightarrow \mathcal{U}(\mathbb{C}^2)$                            | $U : \mathcal{X} \rightarrow \mathcal{U}(\mathbb{C}^2)$                            | $\mathcal{X} = \mathbb{R}^2$ . Suppose that $U : \mathcal{X} \rightarrow \mathcal{U}(\mathbb{C}^2)$ is a quantum unitary    |
| rph8-g15q_F00283 | 17   | ■ 1.000 | $(s, t) \in \mathbb{R}^2$                                                         | $(s, t) \in \mathbb{R}^2$                                                          | for all $(s, t) \in \mathbb{R}^2$ . Then, this defines a perfectly reasonable                                               |
| rph8-g15q_F00284 | 17   | ■ 1.000 | $\mathcal{M}: \mathbb{R}^2 \rightarrow \mathfrak{su}(2)$                          | $\mathcal{M} : \mathbb{R}^2 \rightarrow \mathfrak{su}(2)$                          | sociated with Eq. (A5). First, let $\mathcal{M} : \mathbb{R}^2 \rightarrow \mathfrak{su}(2)$ be the                         |
| rph8-g15q_F00285 | 17   | ■ 1.000 | $\mathcal{M}((1,0))=-i X$                                                         | $\mathcal{M}((1,0)) = -iX$                                                         | linear map uniquely determined by $\mathcal{M}((1,0)) = -iX$ and                                                            |
| rph8-g15q_F00286 | 17   | ■ 1.000 | $\mathcal{M}((0,1))=-i Y$                                                         | $\mathcal{M}((0,1)) = -iY$                                                         | $\mathcal{M}((0,1)) = -iY$ , so that $\mathcal{M}((s,t)) = -isX - itY$ for all                                              |
| rph8-g15q_F00287 | 17   | ■ 1.000 | $\mathcal{M}((s, t))=-i s X-i t Y$                                                | $\mathcal{M}((s,t)) = -isX - itY$                                                  | $\mathcal{M}((0,1)) = -iY$ , so that $\mathcal{M}((s,t)) = -isX - itY$ for all                                              |
| rph8-g15q_F00288 | 17   | ■ 0.968 | $\exp : \mathbb{R}^2 \rightarrow \mathbb{R}^2$                                    | $\exp : \mathbb{R}^2 \rightarrow \mathbb{R}^2$                                     | $(s, t) \in \mathbb{R}^2$ . Since the exponential map $\exp : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ going                  |
| rph8-g15q_F00289 | 17   | ■ 1.000 | $\mathbb{R}^2$                                                                    | $\mathbb{R}^2$                                                                     | from the Lie algebra of the Lie group $\mathbb{R}^2$ (with group struc-                                                     |
| rph8-g15q_F00290 | 17   | ■ 0.995 | $\mathbb{R}_\alpha \subset \mathbb{R}^2$                                          | $\mathbb{R}_\alpha \subset \mathbb{R}^2$                                           | $\mathbb{R}_\alpha \subset \mathbb{R}^2$ generated by some nonzero vector $\alpha$ , i.e.,                                  |
| rph8-g15q_F00291 | 17   | ■ 1.000 | $U _{\mathbb{R}_\alpha}: \mathbb{R}_\alpha \rightarrow \mathcal{U}(\mathbb{C}^2)$ | $U _{\mathbb{R}_\alpha} : \mathbb{R}_\alpha \rightarrow \mathcal{U}(\mathbb{C}^2)$ | group homomorphism $U _{\mathbb{R}_\alpha} : \mathbb{R}_\alpha \rightarrow \mathcal{U}(\mathbb{C}^2)$ since                 |
| rph8-g15q_F00292 | 17   | ■ 1.000 | $W:=\alpha \cdot(X, Y)$                                                           | $W := \alpha \cdot (X, Y)$                                                         | where $W := \alpha \cdot (X, Y)$ is the dot product of $\alpha$ with the vector                                             |
| rph8-g15q_F00293 | 17   | ■ 0.993 | $X, Y$                                                                            | $X, Y$                                                                             | of operators $(X, Y)$ . The second equality in Eq. (A8) holds                                                               |
| rph8-g15q_F00294 | 17   | ■ 1.000 | $W$                                                                               | $W$                                                                                | because every matrix $W$ commutes with itself. In other words,                                                              |
| rph8-g15q_F00295 | 17   | ■ 1.000 | $\mathbb{R} \xrightarrow{\alpha} \mathbb{R}^2$                                    | $\mathbb{R} \xrightarrow{\alpha} \mathbb{R}^2$                                     | the one-parameter subgroup $\mathbb{R} \xrightarrow{\alpha} \mathbb{R}^2$ of $\mathbb{R}^2$ sending $r \in \mathbb{R}$      |
| rph8-g15q_F00296 | 17   | ■ 1.000 | $r \in \mathbb{R}$                                                                | $r \in \mathbb{R}$                                                                 | the one-parameter subgroup $\mathbb{R} \xrightarrow{\alpha} \mathbb{R}^2$ of $\mathbb{R}^2$ sending $r \in \mathbb{R}$      |
| rph8-g15q_F00297 | 17   | ■ 1.000 | $\alpha(r):=r \alpha$                                                             | $\alpha(r) := r\alpha$                                                             | to $\alpha(r) := r\alpha$ is a one-parameter subgroup of $\mathbb{R}^2$ that gets                                           |
| rph8-g15q_F00298 | 17   | ■ 1.000 | $\mathcal{U}(\mathbb{C}^2)$                                                       | $\mathcal{U}(\mathbb{C}^2)$                                                        | of $\mathcal{U}(\mathbb{C}^2)$ that sends $r \in \mathbb{R}$ to $U(r\alpha)$ [126].                                         |
| rph8-g15q_F00299 | 17   | ■ 1.000 | $U(r \alpha)$                                                                     | $U(r\alpha)$                                                                       | of $\mathcal{U}(\mathbb{C}^2)$ that sends $r \in \mathbb{R}$ to $U(r\alpha)$ [126].                                         |
| rph8-g15q_F00300 | 17   | ■ 1.000 | $f: G \rightarrow$                                                                | $f : G \rightarrow$                                                                | whose objects are Lie groups and where a morphism $f : G \rightarrow$                                                       |
| rph8-g15q_F00301 | 17   | ■ 1.000 | $\mathbb{R} \xrightarrow{\gamma} G$                                               | $\mathbb{R} \xrightarrow{\gamma} G$                                                | every one-parameter subgroup $\mathbb{R} \xrightarrow{\gamma} G$ in $G$ , the composite                                     |
| rph8-g15q_F00302 | 17   | ■ 1.000 | $\mathbb{R} \xrightarrow{\gamma} G \xrightarrow{f} H$                             | $\mathbb{R} \xrightarrow{\gamma} G \xrightarrow{f} H$                              | $\mathbb{R} \xrightarrow{\gamma} G \xrightarrow{f} H$ is a one-parameter subgroup in $H$ .                                  |
| rph8-g15q_F00303 | 17   | ■ 1.000 | $G \xrightarrow{f} H$                                                             | $G \xrightarrow{f} H$                                                              | <i>Theorem A2.</i> Let $G \xrightarrow{f} H$ be an OPS homomorphism of                                                      |
| rph8-g15q_F00304 | 17   | ■ 1.000 | $x \in \mathfrak{g}$                                                              | $x \in \mathfrak{g}$                                                               | for all $x \in \mathfrak{g}$ .                                                                                              |
| rph8-g15q_F00305 | 17   | ■ 1.000 | $1_G \in G$                                                                       | $1_G \in G$                                                                        | called the <i>pushforward</i> ) of $f$ at the identity $1_G \in G$ , i.e., $\mathcal{M} :=$                                 |
| rph8-g15q_F00306 | 17   | ■ 1.000 | $\mathcal{M}:=$                                                                   | $\mathcal{M} :=$                                                                   | called the <i>pushforward</i> ) of $f$ at the identity $1_G \in G$ , i.e., $\mathcal{M} :=$                                 |
| rph8-g15q_F00307 | 17   | ■ 1.000 | $Df _{1_G}$                                                                       | $Df _{1_G}$                                                                        | $Df _{1_G}$ . To be somewhat self-contained, let us recall how $\mathcal{M}$                                                |
| rph8-g15q_F00308 | 17   | ■ 1.000 | $\gamma(0)=1_G$                                                                   | $\gamma(0) = 1_G$                                                                  | groups). For any smooth curve $\gamma : \mathbb{R} \rightarrow G$ with $\gamma(0) = 1_G$                                    |
| rph8-g15q_F00309 | 17   | ■ 1.000 | $\left.\frac{d}{dt}\gamma\right _{t=0}=x \in \mathfrak{g}$                        | $\left.\frac{d}{dt}\gamma\right _{t=0} = x \in \mathfrak{g}$                       | and $\frac{d}{dt}\gamma _{t=0} = x \in \mathfrak{g}$ , we have $\frac{d}{dt}(f \circ \gamma) _{t=0} = \mathcal{M}(x)$ . The |

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| rph8-g15q_F00310 | 17   | ■ 1.000 | $\left.\frac{d}{dt}(f \circ \gamma)\right _{t=0} = \mathcal{M}(x)$                                             | $\left.\frac{d}{dt}(f \circ \gamma)\right _{t=0} = \mathcal{M}(x)$                                             | and $\left.\frac{d}{dt}\gamma\right _{t=0} = x \in \mathfrak{g}$ , we have $\left.\frac{d}{dt}(f \circ \gamma)\right _{t=0} = \mathcal{M}(x)$ . The |
| rph8-g15q_F00311 | 17   | ■ 1.000 | $\left.\frac{d}{dt}\gamma\right _{t=0} = x$                                                                    | $\left.\frac{d}{dt}\gamma\right _{t=0} = x$                                                                    | $\left.\frac{d}{dt}\gamma\right _{t=0} = x$ . By Eq. (A2) for $\gamma$ , this means that                                                            |
| rph8-g15q_F00312 | 17   | ■ 1.000 | $f \circ \gamma$                                                                                               | $f \circ \gamma$                                                                                               | by Eq. (A3) for $f \circ \gamma$ , there exists a unique element $y \in \mathfrak{h}$                                                               |
| rph8-g15q_F00313 | 17   | ■ 1.000 | $y \in \mathfrak{h}$                                                                                           | $y \in \mathfrak{h}$                                                                                           | by Eq. (A3) for $f \circ \gamma$ , there exists a unique element $y \in \mathfrak{h}$                                                               |
| rph8-g15q_F00314 | 18   | ■ 1.000 | $G = \mathbb{R}^d$                                                                                             | $G = \mathbb{R}^d$                                                                                             | unitary encodings by taking $G = \mathbb{R}^d$ to be the additive group                                                                             |
| rph8-g15q_F00315 | 18   | ■ 1.000 | $\mathfrak{g} = \mathbb{R}^d$                                                                                  | $\mathfrak{g} = \mathbb{R}^d$                                                                                  | and $\mathfrak{g} = \mathbb{R}^d$ to be its associated Lie algebra. Note that in this                                                               |
| rph8-g15q_F00316 | 18   | ■ 1.000 | $\mathbb{R}^d \xrightarrow{f} G$                                                                               | $\mathbb{R}^d \xrightarrow{f} G$                                                                               | <i>Corollary A1.</i> Let $\mathbb{R}^d \xrightarrow{f} G$ be an OPS homomorphism of                                                                 |
| rph8-g15q_F00317 | 18   | ■ 0.976 | $\mathcal{M} : \mathbb{R}^d \rightarrow$                                                                       | $\mathcal{M} : \mathbb{R}^d \rightarrow$                                                                       | Lie groups. Then, there exists a unique linear map $\mathcal{M} : \mathbb{R}^d \rightarrow$                                                         |
| rph8-g15q_F00318 | 18   | ■ 1.000 | $f : \mathbb{R}^d \rightarrow G$                                                                               | $f : \mathbb{R}^d \rightarrow G$                                                                               | $\mathfrak{g}$ , the function $f : \mathbb{R}^d \rightarrow G$ specified by Eq. (A15) defines an                                                    |
| rph8-g15q_F00319 | 18   | ■ 1.000 | $\mathbb{R}^d \rightarrow \mathcal{U}(\mathbb{C}^n)$                                                           | $\mathbb{R}^d \rightarrow \mathcal{U}(\mathbb{C}^n)$                                                           | $\mathbb{R}^d \rightarrow \mathcal{U}(\mathbb{C}^n)$ of the form                                                                                    |
| rph8-g15q_F00320 | 18   | ■ 0.999 | $\mathcal{L} : \mathbb{R}^d \rightarrow \mathcal{B}(\mathbb{C}^n)$                                             | $\mathcal{L} : \mathbb{R}^d \rightarrow \mathcal{B}(\mathbb{C}^n)$                                             | where $\mathcal{L} : \mathbb{R}^d \rightarrow \mathcal{B}(\mathbb{C}^n)$ is a linear map such that $\mathcal{L}(x)$ is self-                        |
| rph8-g15q_F00321 | 18   | ■ 1.000 | $\mathbb{R}^d \xrightarrow{U} \mathcal{U}(\mathbb{C}^n)$                                                       | $\mathbb{R}^d \xrightarrow{U} \mathcal{U}(\mathbb{C}^n)$                                                       | adjoint for each $x \in \mathbb{R}^d$ , defines a morphism $\mathbb{R}^d \xrightarrow{U} \mathcal{U}(\mathbb{C}^n)$                                 |
| rph8-g15q_F00322 | 18   | ■ 1.000 | $U(x+y) = e^{-i\mathcal{L}(x+y)}$                                                                              | $U(x+y) = e^{-i\mathcal{L}(x+y)}$                                                                              | general. This is because $U(x+y) = e^{-i\mathcal{L}(x+y)}$ is <i>not</i> in gen-                                                                    |
| rph8-g15q_F00323 | 18   | ■ 1.000 | $U(x)U(y) = e^{-i\mathcal{L}(x)}e^{-i\mathcal{L}(y)}$                                                          | $U(x)U(y) = e^{-i\mathcal{L}(x)}e^{-i\mathcal{L}(y)}$                                                          | eral equal to $U(x)U(y) = e^{-i\mathcal{L}(x)}e^{-i\mathcal{L}(y)}$ . Because the group                                                             |
| rph8-g15q_F00324 | 18   | ■ 1.000 | $X \subseteq \mathbb{R}^2$                                                                                     | $X \subseteq \mathbb{R}^2$                                                                                     | a dataset $X \subseteq \mathbb{R}^2$ . The labeling is determined by a binary                                                                       |
| rph8-g15q_F00325 | 18   | ■ 1.000 | $c : \mathcal{X} \rightarrow \{-1, 0, +1\}$                                                                    | $c : \mathcal{X} \rightarrow \{-1, 0, +1\}$                                                                    | <i>classifier</i> , which is a function $c : \mathcal{X} \rightarrow \{-1, 0, +1\}$ , where                                                         |
| rph8-g15q_F00326 | 18   | ■ 1.000 | $O \in$                                                                                                        | $O \in$                                                                                                        | $\rho : \mathcal{X} \rightarrow \mathcal{S}(\mathcal{H})$ if and only if there exists an observable $O \in$                                         |
| rph8-g15q_F00327 | 18   | ■ 0.997 | $O$                                                                                                            | $O$                                                                                                            | $\mathcal{B}(\mathcal{H})$ such that the spectrum of $O$ is $\{-1, +1\}$ and                                                                        |
| rph8-g15q_F00328 | 18   | ■ 0.997 | $\{-1, +1\}$                                                                                                   | $\{-1, +1\}$                                                                                                   | $\mathcal{B}(\mathcal{H})$ such that the spectrum of $O$ is $\{-1, +1\}$ and                                                                        |
| rph8-g15q_F00329 | 18   | ■ 1.000 | $\rho(x)$                                                                                                      | $\rho(x)$                                                                                                      | which is the expectation value of $O$ in the state $\rho(x)$ . The func-                                                                            |
| rph8-g15q_F00330 | 18   | ■ 1.000 | $y$                                                                                                            | $y$                                                                                                            | tional $y$ is called the <i>quantum classifier</i> and the set $y^{-1}(0) \subseteq \mathcal{X}$                                                    |
| rph8-g15q_F00331 | 18   | ■ 1.000 | $y^{-1}(0) \subseteq \mathcal{X}$                                                                              | $y^{-1}(0) \subseteq \mathcal{X}$                                                                              | tional $y$ is called the <i>quantum classifier</i> and the set $y^{-1}(0) \subseteq \mathcal{X}$                                                    |
| rph8-g15q_F00332 | 18   | ■ 1.000 | $x_1, x_2 \in \mathbb{R}$                                                                                      | $x_1, x_2 \in \mathbb{R}$                                                                                      | for all $x_1, x_2 \in \mathbb{R}$ . Note that invariance of the quantum clas-                                                                       |
| rph8-g15q_F00333 | 18   | ■ 1.000 | $ \psi_0\rangle \in \mathcal{H} = \mathbb{C}^4$                                                                | $ \psi_0\rangle \in \mathcal{H} = \mathbb{C}^4$                                                                | Now let $ \psi_0\rangle \in \mathcal{H} = \mathbb{C}^4$ be the initial state                                                                        |
| rph8-g15q_F00334 | 18   | ■ 1.000 | $p = 0.99$                                                                                                     | $p = 0.99$                                                                                                     | with $p = 0.99$ , let $O$ be the observable                                                                                                         |
| rph8-g15q_F00335 | 18   | ■ 0.969 | $(x_1, x_2)$                                                                                                   | $(x_1, x_2)$                                                                                                   | The orange region in Fig. 3 consists of the points $(x_1, x_2)$ such                                                                                |
| rph8-g15q_F00336 | 18   | ■ 1.000 | $y(x_1, x_2) < 0$                                                                                              | $y(x_1, x_2) < 0$                                                                                              | that $y(x_1, x_2) < 0$ , while the blue region consists of the points                                                                               |
| rph8-g15q_F00337 | 18   | ■ 0.999 | $y(x_1, x_2) > 0$                                                                                              | $y(x_1, x_2) > 0$                                                                                              | $(x_1, x_2)$ such that $y(x_1, x_2) > 0$ . The boundary between the                                                                                 |
| rph8-g15q_F00338 | 18   | ■ 0.965 | $y(x_1, x_2) =$                                                                                                | $y(x_1, x_2) =$                                                                                                | two regions consists of the points $(x_1, x_2)$ such that $y(x_1, x_2) =$                                                                           |
| rph8-g15q_F00339 | 18   | ■ 1.000 | $U : \mathbb{R}^2 \rightarrow \mathcal{U}(\mathbb{C}^4)$                                                       | $U : \mathbb{R}^2 \rightarrow \mathcal{U}(\mathbb{C}^4)$                                                       | encodings $U : \mathbb{R}^2 \rightarrow \mathcal{U}(\mathbb{C}^4)$ are of the form $U(x) = e^{-i\mathcal{L}(x)}$                                    |
| rph8-g15q_F00340 | 18   | ■ 1.000 | $U(x) = e^{-i\mathcal{L}(x)}$                                                                                  | $U(x) = e^{-i\mathcal{L}(x)}$                                                                                  | encodings $U : \mathbb{R}^2 \rightarrow \mathcal{U}(\mathbb{C}^4)$ are of the form $U(x) = e^{-i\mathcal{L}(x)}$                                    |
| rph8-g15q_F00341 | 18   | ■ 1.000 | $\mathcal{L} : \mathbb{R}^2 \rightarrow \mathfrak{su}(4)$                                                      | $\mathcal{L} : \mathbb{R}^2 \rightarrow \mathfrak{su}(4)$                                                      | for some linear transformation $\mathcal{L} : \mathbb{R}^2 \rightarrow \mathfrak{su}(4)$ as in Eq. (14).                                            |
| rph8-g15q_F00342 | 18   | ■ 1.000 | $e_1, e_2$                                                                                                     | $e_1, e_2$                                                                                                     | the value of $\mathcal{L}$ on the basis elements $e_1, e_2$ of $\mathbb{R}^2$ . Namely,                                                             |
| rph8-g15q_F00343 | 18   | ■ 1.000 | $L_1 := \mathcal{L}(e_1)$                                                                                      | $L_1 := \mathcal{L}(e_1)$                                                                                      | $L_1 := \mathcal{L}(e_1)$ and $L_2 := \mathcal{L}(e_2)$ . The fact that $L_1$ is a traceless                                                        |
| rph8-g15q_F00344 | 18   | ■ 1.000 | $L_2 := \mathcal{L}(e_2)$                                                                                      | $L_2 := \mathcal{L}(e_2)$                                                                                      | $L_1 := \mathcal{L}(e_1)$ and $L_2 := \mathcal{L}(e_2)$ . The fact that $L_1$ is a traceless                                                        |
| rph8-g15q_F00345 | 18   | ■ 1.000 | $a_{11}, a_{22}, a_{33}, a_{44} \in \mathbb{R}, a_{12}, a_{13}, a_{14}, a_{23}, a_{24}, a_{34} \in \mathbb{C}$ | $a_{11}, a_{22}, a_{33}, a_{44} \in \mathbb{R}, a_{12}, a_{13}, a_{14}, a_{23}, a_{24}, a_{34} \in \mathbb{C}$ | where $a_{11}, a_{22}, a_{33}, a_{44} \in \mathbb{R}, a_{12}, a_{13}, a_{14}, a_{23}, a_{24}, a_{34} \in \mathbb{C}$ ,                              |
| rph8-g15q_F00346 | 18   | ■ 1.000 | $a_{11} + a_{22} + a_{33} + a_{44} = 0$                                                                        | $a_{11} + a_{22} + a_{33} + a_{44} = 0$                                                                        | and $a_{11} + a_{22} + a_{33} + a_{44} = 0$ , and similarly for the matrix                                                                          |
| rph8-g15q_F00347 | 18   | ■ 0.994 | $L_2$                                                                                                          | $L_2$                                                                                                          | $L_2$ . This parametrization leads to a $2(4 + 2(6) - 1) = 30$ -                                                                                    |
| rph8-g15q_F00348 | 18   | ■ 0.994 | $2(4 + 2(6) - 1) = 30$                                                                                         | $2(4 + 2(6) - 1) = 30$                                                                                         | $L_2$ . This parametrization leads to a $2(4 + 2(6) - 1) = 30$ -                                                                                    |
| rph8-g15q_F00349 | 18   | ■ 1.000 | $\mathbb{R}^2 \rightarrow \mathcal{U}(\mathbb{C}^4)$                                                           | $\mathbb{R}^2 \rightarrow \mathcal{U}(\mathbb{C}^4)$                                                           | $\mathbb{R}^2 \rightarrow \mathcal{U}(\mathbb{C}^4)$ that are of the form $U(x) = e^{-i\mathcal{L}(x)}$ for some                                    |
| rph8-g15q_F00350 | 18   | ■ 0.989 | $L_2 =$                                                                                                        | $L_2 =$                                                                                                        | The equivariance constraint $L_2 = \text{SWAP } L_1 \text{ SWAP}$ from Eq.                                                                          |



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| rph8-g15q_F00351 | 19   | ■ 1.000 | $\left\{X \otimes X, L_1\right\}=0$                                     | $\{X \otimes X, L_1\} = 0$                                              | $\{X \otimes X, L_1\} = 0$ from Eq. (18) yields                                                                                |
| rph8-g15q_F00352 | 19   | ■ 1.000 | $\mathbb{R}^n$                                                          | $\mathbb{R}^n$                                                          | cussion on metric spaces that have underlying sets $\mathbb{R}^n$ and                                                          |
| rph8-g15q_F00353 | 19   | ■ 0.936 | $\mathcal{X}=\mathbb{R}^n$                                              | $\mathcal{X} = \mathbb{R}^n$                                            | In more detail, let $\mathcal{X} = \mathbb{R}^n$ and equip it with a Mahalanobis                                               |
| rph8-g15q_F00354 | 19   | ■ 0.432 | $\left(\mathcal{Y}=\mathbb{R}^m, d_{\mathcal{Y}}\right)$                | $(\mathcal{Y} = \mathbb{R}^m, d_{\mathcal{Y}})$                         | distance metric, where $(\mathcal{Y} = \mathbb{R}^m, d_{\mathcal{Y}})$ is a Euclidean space                                    |
| rph8-g15q_F00355 | 19   | ■ 1.000 | $\mathbb{R}^n \xrightarrow{\varphi} \mathbb{R}^m$                       | $\mathbb{R}^n \xrightarrow{\varphi} \mathbb{R}^m$                       | (so that $d_{\mathcal{Y}}$ is the Euclidean distance) and $\mathbb{R}^n \xrightarrow{\varphi} \mathbb{R}^m$ is an              |
| rph8-g15q_F00356 | 19   | ■ 1.000 | $m \times n$                                                            | $m \times n$                                                            | injective linear map, described by some $m \times n$ matrix $V$ with                                                           |
| rph8-g15q_F00357 | 19   | ■ 1.000 | $V$                                                                     | $V$                                                                     | injective linear map, described by some $m \times n$ matrix $V$ with                                                           |
| rph8-g15q_F00358 | 19   | ■ 1.000 | $V^T$                                                                   | $V^T$                                                                   | where $V^T$ denotes the transpose of $V$ . Although this exam-                                                                 |
| rph8-g15q_F00359 | 19   | ■ 0.871 | $\left(\mathbb{R}^m, A\right)$                                          | $(\mathbb{R}^m, A)$                                                     | $(\mathbb{R}^m, A)$ , with $m \in \mathbb{N}$ and $A$ a positive-definite $m \times m$ matrix.                                 |
| rph8-g15q_F00360 | 19   | ■ 0.871 | $m \in \mathbb{N}$                                                      | $m \in \mathbb{N}$                                                      | $(\mathbb{R}^m, A)$ , with $m \in \mathbb{N}$ and $A$ a positive-definite $m \times m$ matrix.                                 |
| rph8-g15q_F00361 | 19   | ■ 0.871 | $A$                                                                     | $A$                                                                     | $(\mathbb{R}^m, A)$ , with $m \in \mathbb{N}$ and $A$ a positive-definite $m \times m$ matrix.                                 |
| rph8-g15q_F00362 | 19   | ■ 0.871 | $m \times m$                                                            | $m \times m$                                                            | $(\mathbb{R}^m, A)$ , with $m \in \mathbb{N}$ and $A$ a positive-definite $m \times m$ matrix.                                 |
| rph8-g15q_F00363 | 19   | ■ 0.945 | $\mathbb{R}^m, A$                                                       | $\mathbb{R}^m, A$                                                       | A morphism in <b>Mahal</b> from $(\mathbb{R}^m, A)$ to $(\mathbb{R}^n, B)$ is an $n \times m$                                  |
| rph8-g15q_F00364 | 19   | ■ 0.945 | $\mathbb{R}^n, B$                                                       | $\mathbb{R}^n, B$                                                       | A morphism in <b>Mahal</b> from $(\mathbb{R}^m, A)$ to $(\mathbb{R}^n, B)$ is an $n \times m$                                  |
| rph8-g15q_F00365 | 19   | ■ 0.945 | $n \times m$                                                            | $n \times m$                                                            | A morphism in <b>Mahal</b> from $(\mathbb{R}^m, A)$ to $(\mathbb{R}^n, B)$ is an $n \times m$                                  |
| rph8-g15q_F00366 | 19   | ■ 1.000 | $V^T B V=A$                                                             | $V^T B V = A$                                                           | matrix $V$ with trivial kernel such that $V^T B V = A$ . The trivial                                                           |
| rph8-g15q_F00367 | 19   | ■ 1.000 | $\mathbb{R}^m$                                                          | $\mathbb{R}^m$                                                          | $\mathbb{R}^m$ to $\mathbb{R}^n$ . The category <b>Mahal</b> is <i>equivalent</i> [130] to the                                 |
| rph8-g15q_F00368 | 19   | ■ 1.000 | $\mathbb{R}^m \xrightarrow{V} \mathbb{R}^n$                             | $\mathbb{R}^m \xrightarrow{V} \mathbb{R}^n$                             | the task can be described as finding a linear map $\mathbb{R}^m \xrightarrow{V} \mathbb{R}^n$ ,                                |
| rph8-g15q_F00369 | 19   | ■ 1.000 | $y \in \mathcal{Y}$                                                     | $y \in \mathcal{Y}$                                                     | holds for all $y \in \mathcal{Y}$ . In particular, for any $x_3 \in \mathcal{X}$ ,                                             |
| rph8-g15q_F00370 | 19   | ■ 1.000 | $x_3 \in \mathcal{X}$                                                   | $x_3 \in \mathcal{X}$                                                   | holds for all $y \in \mathcal{Y}$ . In particular, for any $x_3 \in \mathcal{X}$ ,                                             |
| rph8-g15q_F00371 | 19   | ■ 1.000 | $d_{\mathcal{X}}(x, x)=0$                                               | $d_{\mathcal{X}}(x, x) = 0$                                             | true that $d_{\mathcal{X}}(x, x) = 0$ for all $x \in \mathcal{X}$ .                                                            |
| rph8-g15q_F00372 | 19   | ■ 1.000 | $f\left(x_1\right)=f\left(x_2\right)$                                   | $f\left(x_1\right)=f\left(x_2\right)$                                   | then $f\left(x_1\right)=f\left(x_2\right)$ . Hence, $f$ is one to one if and only                                              |
| rph8-g15q_F00373 | 19   | ■ 1.000 | $x_1, x_2$                                                              | $x_1, x_2$                                                              | if $d_{\mathcal{X}}\left(x_1, x_2\right)=0$ implies $x_1=x_2$ for all $x_1, x_2$ that satisfy                                  |
| rph8-g15q_F00374 | 19   | ■ 1.000 | $X \subset \mathcal{X}$                                                 | $X \subset \mathcal{X}$                                                 | classes. Although the given data $X \subset \mathcal{X}$ shown in Fig. 8 are                                                   |
| rph8-g15q_F00375 | 20   | ■ 1.000 | $\{A, B\}$                                                              | $\{A, B\}$                                                              | Let $\{A, B\}$ be a partition of $X$ , where the elements of $A$ are                                                           |
| rph8-g15q_F00376 | 20   | ■ 0.997 | $B$                                                                     | $B$                                                                     | labeled as class +1 and the elements of $B$ are labeled as class                                                               |
| rph8-g15q_F00377 | 20   | ■ 0.954 | $c: \mathcal{X} \rightarrow\{-1,0,1\}$                                  | $c: \mathcal{X} \rightarrow\{-1,0,1\}$                                  | −1. The associated classifier $c: \mathcal{X} \rightarrow\{-1,0,1\}$ satisfies                                                 |
| rph8-g15q_F00378 | 20   | ■ 0.965 | $\# A$                                                                  | $\# A$                                                                  | where $\# A$ and $\# B$ denote the cardinalities of $A$ and $B$ , respec-                                                      |
| rph8-g15q_F00379 | 20   | ■ 0.965 | $\# B$                                                                  | $\# B$                                                                  | where $\# A$ and $\# B$ denote the cardinalities of $A$ and $B$ , respec-                                                      |
| rph8-g15q_F00380 | 20   | ■ 1.000 | $\mathcal{X} \rightarrow \mathcal{S}(\mathcal{H})$                      | $\mathcal{X} \rightarrow \mathcal{S}(\mathcal{H})$                      | not a trivial task because the space of encodings $\mathcal{X} \rightarrow \mathcal{S}(\mathcal{H})$                           |
| rph8-g15q_F00381 | 20   | ■ 1.000 | $\rho: \Theta \times \mathcal{X} \rightarrow \mathcal{S}(\mathcal{H})$  | $\rho: \Theta \times \mathcal{X} \rightarrow \mathcal{S}(\mathcal{H})$  | a <i>parametrized</i> family of encodings $\rho: \Theta \times \mathcal{X} \rightarrow \mathcal{S}(\mathcal{H})$ ,             |
| rph8-g15q_F00382 | 20   | ■ 1.000 | $\Theta=\left[0,2 \pi\right]^4$                                         | $\Theta=[0,2 \pi]^4$                                                    | this work.) More specifically, we take $\Theta=[0,2 \pi]^4$ and write                                                          |
| rph8-g15q_F00383 | 20   | ■ 1.000 | $\theta=\left(\theta_1, \theta_2, \theta_3, \theta_4\right) \in \Theta$ | $\theta=\left(\theta_1, \theta_2, \theta_3, \theta_4\right) \in \Theta$ | $\theta=\left(\theta_1, \theta_2, \theta_3, \theta_4\right) \in \Theta$ and $\rho_{\theta}(x):=\rho(\theta, x)$ . Moreover, we |
| rph8-g15q_F00384 | 20   | ■ 1.000 | $\rho_{\theta}(x):=\rho(\theta, x)$                                     | $\rho_{\theta}(x):=\rho(\theta, x)$                                     | $\theta=\left(\theta_1, \theta_2, \theta_3, \theta_4\right) \in \Theta$ and $\rho_{\theta}(x):=\rho(\theta, x)$ . Moreover, we |
| rph8-g15q_F00385 | 20   | ■ 1.000 | $U: \Theta \times \mathcal{X} \rightarrow \mathcal{U}(\mathcal{H})$     | $U: \Theta \times \mathcal{X} \rightarrow \mathcal{U}(\mathcal{H})$     | $U: \Theta \times \mathcal{X} \rightarrow \mathcal{U}(\mathcal{H})$ , where                                                    |
| rph8-g15q_F00386 | 20   | ■ 1.000 | $\rho_A$                                                                | $\rho_A$                                                                | which is an observable that depends on $\theta$ (since $\rho_A$ and $\rho_B$                                                   |
| rph8-g15q_F00387 | 20   | ■ 1.000 | $\rho_B$                                                                | $\rho_B$                                                                | which is an observable that depends on $\theta$ (since $\rho_A$ and $\rho_B$                                                   |
| rph8-g15q_F00388 | 20   | ■ 0.999 | $\lambda \in(0,1]$                                                      | $\lambda \in(0,1]$                                                      | matrices. Then, there exists a $\lambda \in(0,1]$ such that the eigen-                                                         |
| rph8-g15q_F00389 | 20   | ■ 1.000 | $\rho_A-\rho_B$                                                         | $\rho_A-\rho_B$                                                         | values of $\rho_A-\rho_B$ are $\{-\lambda,+\lambda\}$ .                                                                        |
| rph8-g15q_F00390 | 20   | ■ 1.000 | $\{-\lambda,+\lambda\}$                                                 | $\{-\lambda,+\lambda\}$                                                 | values of $\rho_A-\rho_B$ are $\{-\lambda,+\lambda\}$ .                                                                        |
| rph8-g15q_F00391 | 20   | ■ 1.000 | $\lambda_1$                                                             | $\lambda_1$                                                             | $\lambda_1$ and $\lambda_2$ denote the two eigenvalues of $\rho_A-\rho_B$ . Since the                                          |
| rph8-g15q_F00392 | 20   | ■ 1.000 | $\lambda_2$                                                             | $\lambda_2$                                                             | $\lambda_1$ and $\lambda_2$ denote the two eigenvalues of $\rho_A-\rho_B$ . Since the                                          |
| rph8-g15q_F00393 | 20   | ■ 0.955 | $\rho_A \neq \rho_B$                                                    | $\rho_A \neq \rho_B$                                                    | of $\rho_A-\rho_B$ are between −1 and 1. Since $\rho_A \neq \rho_B$ , at least one                                             |
| rph8-g15q_F00394 | 20   | ■ 0.982 | $\lambda_1+\lambda_2=0$                                                 | $\lambda_1+\lambda_2=0$                                                 | i.e., $\lambda_1+\lambda_2=0$ . In other words, $\lambda_2=-\lambda_1$ . Setting $\lambda=\left \lambda_1\right $              |
| rph8-g15q_F00395 | 20   | ■ 0.982 | $\lambda_2=-\lambda_1$                                                  | $\lambda_2=-\lambda_1$                                                  | i.e., $\lambda_1+\lambda_2=0$ . In other words, $\lambda_2=-\lambda_1$ . Setting $\lambda=\left \lambda_1\right $              |
| rph8-g15q_F00396 | 20   | ■ 0.982 | $\lambda=\left \lambda_1\right $                                        | $\lambda=\left \lambda_1\right $                                        | i.e., $\lambda_1+\lambda_2=0$ . In other words, $\lambda_2=-\lambda_1$ . Setting $\lambda=\left \lambda_1\right $              |
| rph8-g15q_F00397 | 20   | ■ 1.000 | $A \cup B=X$                                                            | $A \cup B=X$                                                            | gether with a partition $A \cup B=X$ , with elements of $A$ and $B$                                                            |
| rph8-g15q_F00398 | 20   | ■ 1.000 | $O=\rho_A-\rho_B$                                                       | $O=\rho_A-\rho_B$                                                       | Eq. (D6), are distinct density matrices. Set $O=\rho_A-\rho_B$ and                                                             |
| rph8-g15q_F00399 | 20   | ■ 1.000 | $\lambda>0$                                                             | $\lambda>0$                                                             | let $\{-\lambda,+\lambda\}$ denote the set of eigenvalues of $O$ , with $\lambda>0$ .                                          |

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| rph8-g15q_F00400 | 20   | ■ 1.000 | c                                                                                     | $c$                                                                                     | Finally, let $c$ be the classifier in Eq. (D5) and let $y : \mathcal{X} \rightarrow \mathbb{R}$                                                      |
| rph8-g15q_F00401 | 20   | ■ 0.856 | $x \in X$                                                                             | $x \in X$                                                                               | for all $x \in X$ , where $\text{sgn} : (-\infty, 0) \cup (0, \infty) \rightarrow \{-1, +1\}$ de-                                                    |
| rph8-g15q_F00402 | 20   | ■ 0.856 | $\operatorname{sgn}:(-\infty, 0) \cup (0, \infty) \rightarrow \{-1, +1\}$             | $\text{sgn} : (-\infty, 0) \cup (0, \infty) \rightarrow \{-1, +1\}$                     | for all $x \in X$ , where $\text{sgn} : (-\infty, 0) \cup (0, \infty) \rightarrow \{-1, +1\}$ de-                                                    |
| rph8-g15q_F00403 | 20   | ■ 1.000 | $\operatorname{sgn}(x)=\frac{x}{ x }$                                                 | $\text{sgn}(x) = \frac{x}{ x }$                                                         | notes the sign function $\text{sgn}(x) = \frac{x}{ x }$ .                                                                                            |
| rph8-g15q_F00404 | 20   | ■ 1.000 | $\Pi_+-\Pi_-$                                                                         | $\Pi_+ - \Pi_-$                                                                         | observable $\Pi_+ - \Pi_-$ , where $\Pi_+$ and $\Pi_-$ are the projections                                                                           |
| rph8-g15q_F00405 | 20   | ■ 1.000 | $\Pi_+$                                                                               | $\Pi_+$                                                                                 | observable $\Pi_+ - \Pi_-$ , where $\Pi_+$ and $\Pi_-$ are the projections                                                                           |
| rph8-g15q_F00406 | 20   | ■ 1.000 | $\Pi_-$                                                                               | $\Pi_-$                                                                                 | observable $\Pi_+ - \Pi_-$ , where $\Pi_+$ and $\Pi_-$ are the projections                                                                           |
| rph8-g15q_F00407 | 20   | ■ 1.000 | $\theta_1, \theta_2, \theta_3$                                                        | $\theta_1, \theta_2, \theta_3$                                                          | we obtain specific values of $\theta_1, \theta_2, \theta_3$ , and $\theta_4$ for the                                                                 |
| rph8-g15q_F00408 | 20   | ■ 1.000 | $\theta_4$                                                                            | $\theta_4$                                                                              | we obtain specific values of $\theta_1, \theta_2, \theta_3$ , and $\theta_4$ for the                                                                 |
| rph8-g15q_F00409 | 20   | ■ 1.000 | $\rho:\Theta\times\mathcal{X}\rightarrow\mathscr{S}(\mathbb{C}^2)$                    | $\rho : \Theta \times \mathcal{X} \rightarrow \mathscr{S}(\mathbb{C}^2)$                | parametrized encoding $\rho : \Theta \times \mathcal{X} \rightarrow \mathscr{S}(\mathbb{C}^2)$ . The associated                                      |
| rph8-g15q_F00410 | 20   | ■ 1.000 | $\mathcal{X}\times\mathcal{X}$                                                        | $\mathcal{X} \times \mathcal{X}$                                                        | the distance between points on a discretized grid from $\mathcal{X} \times \mathcal{X}$                                                              |
| rph8-g15q_F00411 | 21   | ■ 1.000 | $G:\mathbf{D}\rightarrow\mathbf{E}$                                                   | $G : \mathbf{D} \rightarrow \mathbf{E}$                                                 | <i>Definition E1.</i> Let $F : \mathbf{C} \rightarrow \mathbf{D}$ and $G : \mathbf{D} \rightarrow \mathbf{E}$ be two                                 |
| rph8-g15q_F00412 | 21   | ■ 1.000 | $G\circ F:\mathbf{C}\rightarrow\mathbf{E}$                                            | $G \circ F : \mathbf{C} \rightarrow \mathbf{E}$                                         | functors. The <i>composite</i> $G \circ F : \mathbf{C} \rightarrow \mathbf{E}$ is the functor that                                                   |
| rph8-g15q_F00413 | 21   | ■ 1.000 | $G(F(\mathcal{X}))$                                                                   | $G(F(\mathcal{X}))$                                                                     | sends each object $\mathcal{X}$ in $\mathbf{C}$ to $G(F(\mathcal{X}))$ and sends each mor-                                                           |
| rph8-g15q_F00414 | 21   | ■ 0.999 | $G(F(f))$                                                                             | $G(F(f))$                                                                               | phism $f : \mathcal{X} \rightarrow \mathcal{Y}$ in $\mathbf{C}$ to the morphism $G(F(f))$ .                                                          |
| rph8-g15q_F00415 | 21   | ■ 0.989 | $F, G$                                                                                | $F, G$                                                                                  | <i>Definition E2.</i> Let $\mathbf{C}$ and $\mathbf{D}$ be categories, and let $F, G :$                                                              |
| rph8-g15q_F00416 | 21   | ■ 0.996 | $\eta$                                                                                | $\eta$                                                                                  | $\mathbf{C} \rightarrow \mathbf{D}$ be two functors. A <i>natural transformation</i> $\eta$ from $F$                                                 |
| rph8-g15q_F00417 | 21   | ■ 1.000 | $\eta:F\Rightarrow G$                                                                 | $\eta : F \Rightarrow G$                                                                | to $G$ , written $\eta : F \Rightarrow G$ , associates with each object $\mathcal{X}$ in $\mathbf{C}$                                                |
| rph8-g15q_F00418 | 21   | ■ 0.948 | $\eta_{\mathcal{X}}:F(\mathcal{X})\rightarrow G(\mathcal{X})$                         | $\eta_{\mathcal{X}} : F(\mathcal{X}) \rightarrow G(\mathcal{X})$                        | a morphism $\eta_{\mathcal{X}} : F(\mathcal{X}) \rightarrow G(\mathcal{X})$ in $\mathbf{D}$ such that the diagram                                    |
| rph8-g15q_F00419 | 21   | ■ 0.996 | $G(f)\circ\eta_{\mathcal{X}}=\eta_{\mathcal{Y}}\circ F(f)$                            | $G(f) \circ \eta_{\mathcal{X}} = \eta_{\mathcal{Y}} \circ F(f)$                         | in $\mathbf{D}$ commutes, i.e., $G(f) \circ \eta_{\mathcal{X}} = \eta_{\mathcal{Y}} \circ F(f)$ , for every mor-                                     |
| rph8-g15q_F00420 | 21   | ■ 1.000 | $\mathcal{C}$                                                                         | $\mathcal{C}$                                                                           | phism $\mathcal{X} \xrightarrow{f} \mathcal{Y}$ in $\mathcal{C}$ . The commutativity of diagram (E1) is                                              |
| rph8-g15q_F00421 | 21   | ■ 1.000 | $\mathbb{C}$                                                                          | $\mathbb{C}$                                                                            | phisms are $\mathbb{C}$ -linear transformations. Moreover, there is a                                                                                |
| rph8-g15q_F00422 | 21   | ■ 1.000 | $(\mathbb{C}_{\text{fs}}^{\mathcal{X}}, 0, +, \cdot)$                                 | $(\mathbb{C}_{\text{fs}}^{\mathcal{X}}, 0, +, \cdot)$                                   | a set $\mathcal{X}$ the vector space $(\mathbb{C}_{\text{fs}}^{\mathcal{X}}, 0, +, \cdot)$ generated by it, which                                    |
| rph8-g15q_F00423 | 21   | ■ 1.000 | $\mathbb{C}_{\text{fs}}^{\mathcal{X}}\rightarrow\mathbb{C}_{\text{fs}}^{\mathcal{Y}}$ | $\mathbb{C}_{\text{fs}}^{\mathcal{X}} \rightarrow \mathbb{C}_{\text{fs}}^{\mathcal{Y}}$ | $\mathcal{Y}$ a linear transformation $\mathbb{C}_{\text{fs}}^{\mathcal{X}} \rightarrow \mathbb{C}_{\text{fs}}^{\mathcal{Y}}$ . This passage defines |
| rph8-g15q_F00424 | 21   | ■ 0.995 | $v$                                                                                   | $v$                                                                                     | $\mathcal{X}$ . Namely, each vector $v$ in $\mathbb{C}_{\text{fs}}^{\mathcal{X}}$ is by definition a function                                        |
| rph8-g15q_F00425 | 21   | ■ 0.995 | $\mathbb{C}_{\text{fs}}^{\mathcal{X}}$                                                | $\mathbb{C}_{\text{fs}}^{\mathcal{X}}$                                                  | $\mathcal{X}$ . Namely, each vector $v$ in $\mathbb{C}_{\text{fs}}^{\mathcal{X}}$ is by definition a function                                        |
| rph8-g15q_F00426 | 21   | ■ 1.000 | $v:\mathcal{X}\rightarrow\mathbb{C}$                                                  | $v : \mathcal{X} \rightarrow \mathbb{C}$                                                | $v : \mathcal{X} \rightarrow \mathbb{C}$ such that $v(x) = 0$ for all but finitely many $x \in \mathcal{X}$ .                                        |
| rph8-g15q_F00427 | 21   | ■ 1.000 | $v(x)=0$                                                                              | $v(x) = 0$                                                                              | $v : \mathcal{X} \rightarrow \mathbb{C}$ such that $v(x) = 0$ for all but finitely many $x \in \mathcal{X}$ .                                        |
| rph8-g15q_F00428 | 21   | ■ 0.999 | $v, w \in \mathbb{C}_{\text{fs}}^{\mathcal{X}}$                                       | $v, w \in \mathbb{C}_{\text{fs}}^{\mathcal{X}}$                                         | signs 0 to every $x \in \mathcal{X}$ . The sum of two functions $v, w \in \mathbb{C}_{\text{fs}}^{\mathcal{X}}$                                      |
| rph8-g15q_F00429 | 21   | ■ 1.000 | $v+w$                                                                                 | $v + w$                                                                                 | is the function $v + w$ whose value on $x \in \mathcal{X}$ is $(v + w)(x) :=$                                                                        |
| rph8-g15q_F00430 | 21   | ■ 1.000 | $(v+w)(x):=$                                                                          | $(v + w)(x) :=$                                                                         | is the function $v + w$ whose value on $x \in \mathcal{X}$ is $(v + w)(x) :=$                                                                        |
| rph8-g15q_F00431 | 21   | ■ 1.000 | $v(x)+w(x)$                                                                           | $v(x) + w(x)$                                                                           | $v(x) + w(x)$ . Meanwhile, if $v$ is such a function and $\lambda \in \mathbb{C}$                                                                    |
| rph8-g15q_F00432 | 21   | ■ 1.000 | $\lambda \in \mathbb{C}$                                                              | $\lambda \in \mathbb{C}$                                                                | $v(x) + w(x)$ . Meanwhile, if $v$ is such a function and $\lambda \in \mathbb{C}$                                                                    |
| rph8-g15q_F00433 | 21   | ■ 1.000 | $\lambda \cdot v$                                                                     | $\lambda \cdot v$                                                                       | is a scalar, then $\lambda \cdot v$ , written as $\lambda v$ for short, is the function                                                              |
| rph8-g15q_F00434 | 21   | ■ 1.000 | $\lambda v$                                                                           | $\lambda v$                                                                             | is a scalar, then $\lambda \cdot v$ , written as $\lambda v$ for short, is the function                                                              |
| rph8-g15q_F00435 | 21   | ■ 1.000 | $\lambda v(x)$                                                                        | $\lambda v(x)$                                                                          | whose value on $x \in \mathcal{X}$ is $\lambda v(x)$ .                                                                                               |
| rph8-g15q_F00436 | 21   | ■ 1.000 | $\delta_x$                                                                            | $\delta_x$                                                                              | The elements $x \in \mathcal{X}$ define a basis $\delta_x$ for $\mathbb{C}_{\text{fs}}^{\mathcal{X}}$ , where the                                    |
| rph8-g15q_F00437 | 21   | ■ 0.998 | $\delta_x(x')=\delta_{xx'}$                                                           | $\delta_x(x') = \delta_{xx'}$                                                           | function $\delta_x$ acts by $\delta_x(x') = \delta_{xx'}$ that is 1 if $x' = x$ and 0 if $x' \neq$                                                   |
| rph8-g15q_F00438 | 21   | ■ 0.998 | $x'=x$                                                                                | $x' = x$                                                                                | function $\delta_x$ acts by $\delta_x(x') = \delta_{xx'}$ that is 1 if $x' = x$ and 0 if $x' \neq$                                                   |
| rph8-g15q_F00439 | 21   | ■ 0.998 | $x'\neq$                                                                              | $x' \neq$                                                                               | function $\delta_x$ acts by $\delta_x(x') = \delta_{xx'}$ that is 1 if $x' = x$ and 0 if $x' \neq$                                                   |
| rph8-g15q_F00440 | 21   | ■ 1.000 | $ x\rangle$                                                                           | $ x\rangle$                                                                             | could write $ x\rangle$ so that $\langle x' x\rangle = \delta_{xx'}$ , but that might be consid-                                                     |
| rph8-g15q_F00441 | 21   | ■ 1.000 | $\langle x' x\rangle=\delta_{xx'}$                                                    | $\langle x'   x \rangle = \delta_{xx'}$                                                 | could write $ x\rangle$ so that $\langle x' x\rangle = \delta_{xx'}$ , but that might be consid-                                                     |
| rph8-g15q_F00442 | 21   | ■ 1.000 | $\mathcal{X}=\mathbb{Z}_2^n=$                                                         | $\mathcal{X} = \mathbb{Z}_2^n =$                                                        | familiar to the quantum information theorist. Let $\mathcal{X} = \mathbb{Z}_2^n =$                                                                   |
| rph8-g15q_F00443 | 21   | ■ 1.000 | $\{0,1\}^n$                                                                           | $\{0, 1\}^n$                                                                            | $\{0, 1\}^n$ be the set of all arrays $x = (x_0, x_1, \dots, x_{n-1})$ whose                                                                         |



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| rph8-g15q_F00444 | 21   | ■ 1.000 | $x=\left(x_{\{0\}}, x_{\{1\}}, \ldots, x_{\{n-1\}}\right)$                                             | $x = (x_0, x_1, \dots, x_{n-1})$                                                                  | $\{0, 1\}^n$ be the set of all arrays $x = (x_0, x_1, \dots, x_{n-1})$ whose                                                                                                 |
| rph8-g15q_F00445 | 21   | ■ 1.000 | $\left\{0,1,2, \ldots, 2^{\{n\}}-1\right\}$                                                            | $\{0, 1, 2, \dots, 2^n - 1\}$                                                                     | representation of a number $\{0, 1, 2, \dots, 2^n - 1\}$ . Then, the                                                                                                         |
| rph8-g15q_F00446 | 21   | ■ 1.000 | $\mathbb{C}^{\mathcal{X}}$                                                                             | $\mathbb{C}^{\mathcal{X}}$                                                                        | vector space $\mathbb{C}^{\mathcal{X}}$ generated by $\mathcal{X}$ is precisely (i.e., naturally                                                                             |
| rph8-g15q_F00447 | 21   | ■ 1.000 | $2^{\{n\}}$                                                                                            | $2^n$                                                                                             | is a finite set with $2^n$ elements.) From this perspective, each                                                                                                            |
| rph8-g15q_F00448 | 21   | ■ 1.000 | $L_{\{f\}}: \mathbb{C}_{\mathrm{fs}}^{\mathcal{X}} \rightarrow$                                        | $L_f : \mathbb{C}_{\mathrm{fs}}^{\mathcal{X}} \rightarrow$                                        | this function, define the linear transformation $L_f : \mathbb{C}_{\mathrm{fs}}^{\mathcal{X}} \rightarrow$                                                                   |
| rph8-g15q_F00449 | 21   | ■ 0.995 | $\mathbb{C}_{\mathrm{fs}}^{\mathcal{Y}}$                                                               | $\mathbb{C}_{\mathrm{fs}}^{\mathcal{Y}}$                                                          | $\mathbb{C}_{\mathrm{fs}}^{\mathcal{Y}}$ as follows. By writing the basis elements of $\mathbb{C}_{\mathrm{fs}}^{\mathcal{X}}$ and                                           |
| rph8-g15q_F00450 | 21   | ■ 0.766 | $ y\rangle$                                                                                            | $ y\rangle$                                                                                       | $\mathbb{C}_{\mathrm{fs}}^{\mathcal{Y}}$ as $ x\rangle$ and $ y\rangle$ , respectively, the linear transformation $L_f$                                                      |
| rph8-g15q_F00451 | 21   | ■ 0.766 | $L_{\{f\}}$                                                                                            | $L_f$                                                                                             | $\mathbb{C}_{\mathrm{fs}}^{\mathcal{Y}}$ as $ x\rangle$ and $ y\rangle$ , respectively, the linear transformation $L_f$                                                      |
| rph8-g15q_F00452 | 21   | ■ 1.000 | $ f(x)\rangle$                                                                                         | $ f(x)\rangle$                                                                                    | sends each basis vector $ x\rangle$ to $ f(x)\rangle$ , which is another ba-                                                                                                 |
| rph8-g15q_F00453 | 21   | ■ 1.000 | $\mathbb{C}^{\mathcal{Y}}$                                                                             | $\mathbb{C}^{\mathcal{Y}}$                                                                        | sis vector in $\mathbb{C}^{\mathcal{Y}}$ . Since a linear transformation is uniquely                                                                                         |
| rph8-g15q_F00454 | 21   | ■ 1.000 | $L_{\{f\}}: \mathbb{C}_{\mathrm{fs}}^{\mathcal{X}} \rightarrow \mathbb{C}_{\mathrm{fs}}^{\mathcal{Y}}$ | $L_f : \mathbb{C}_{\mathrm{fs}}^{\mathcal{X}} \rightarrow \mathbb{C}_{\mathrm{fs}}^{\mathcal{Y}}$ | transforming $f : \mathcal{X} \rightarrow \mathcal{Y}$ to $L_f : \mathbb{C}_{\mathrm{fs}}^{\mathcal{X}} \rightarrow \mathbb{C}_{\mathrm{fs}}^{\mathcal{Y}}$ is often used in |
| rph8-g15q_F00455 | 21   | ■ 0.871 | $(V, 0, +, \cdot)$                                                                                     | $(V, 0, +, \cdot)$                                                                                | going the other way that takes a vector space $(V, 0, +, \cdot)$ and                                                                                                         |
| rph8-g15q_F00456 | 22   | ■ 1.000 | $F(\mathcal{X})=$                                                                                      | $F(\mathcal{X}) =$                                                                                | ing a set $\mathcal{X}$ , constructing the associated vector space $F(\mathcal{X}) =$                                                                                        |
| rph8-g15q_F00457 | 22   | ■ 0.562 | $G(F(\mathcal{X}))=\mathbb{C}_{\mathrm{fs}}^{\mathcal{X}}$                                             | $G(F(\mathcal{X})) = \mathbb{C}_{\mathrm{fs}}^{\mathcal{X}}$                                      | gives a <i>set</i> $G(F(\mathcal{X})) = \mathbb{C}_{\mathrm{fs}}^{\mathcal{X}}$ .                                                                                            |
| rph8-g15q_F00458 | 22   | ■ 1.000 | $G \circ F$                                                                                            | $G \circ F$                                                                                       | hand, we have $G \circ F$ , which we just described. On the other                                                                                                            |
| rph8-g15q_F00459 | 22   | ■ 0.977 | $\mathrm{id}_{\mathrm{Set}}$                                                                           | $\mathrm{id}_{\mathrm{Set}}$                                                                      | hand, we also have the functor $\mathrm{id}_{\mathrm{Set}} : \mathbf{Set} \rightarrow \mathbf{Set}$ that acts                                                                |
| rph8-g15q_F00460 | 22   | ■ 0.955 | $\eta: \mathrm{id}_{\mathrm{Set}} \Rightarrow$                                                         | $\eta : \mathrm{id}_{\mathrm{Set}} \Rightarrow$                                                   | realize bit encoding as the natural transformation $\eta : \mathrm{id}_{\mathrm{Set}} \Rightarrow$                                                                           |
| rph8-g15q_F00461 | 22   | ■ 0.877 | $\eta_{\mathcal{X}}: \mathrm{id}_{\mathrm{Set}}(\mathcal{X}) \rightarrow G(F(\mathcal{X}))$            | $\eta_{\mathcal{X}} : \mathrm{id}_{\mathrm{Set}}(\mathcal{X}) \rightarrow G(F(\mathcal{X}))$      | $\eta_{\mathcal{X}} : \mathrm{id}_{\mathrm{Set}}(\mathcal{X}) \rightarrow G(F(\mathcal{X}))$ , which is a function of the form                                               |
| rph8-g15q_F00462 | 22   | ■ 1.000 | $\eta_{\mathcal{X}}: \mathcal{X} \rightarrow \mathbb{C}_{\mathrm{fs}}^{\mathcal{X}}$                   | $\eta_{\mathcal{X}} : \mathcal{X} \rightarrow \mathbb{C}_{\mathrm{fs}}^{\mathcal{X}}$             | $\eta_{\mathcal{X}} : \mathcal{X} \rightarrow \mathbb{C}_{\mathrm{fs}}^{\mathcal{X}}$ . The definition of this function is precisely                                         |
| rph8-g15q_F00463 | 22   | ■ 1.000 | $\eta_{\mathcal{X}}(x)= x\rangle \in \mathbb{C}_{\mathrm{fs}}^{\mathcal{X}}$                           | $\eta_{\mathcal{X}}(x) =  x\rangle \in \mathbb{C}_{\mathrm{fs}}^{\mathcal{X}}$                    | the vector $\eta_{\mathcal{X}}(x) =  x\rangle \in \mathbb{C}_{\mathrm{fs}}^{\mathcal{X}}$ . So what does the naturality                                                      |
| rph8-g15q_F00464 | 22   | ■ 1.000 | $\{ x\rangle\}$                                                                                        | $\{ x\rangle\}$                                                                                   | how it acts on the basis $\{ x\rangle\}$ . Naturality of $\eta$ then says that                                                                                               |
| rph8-g15q_F00465 | 22   | ■ 1.000 | $\eta_{\mathcal{Y}}(f(x))=L_f(\eta_{\mathcal{X}}(x))$                                                  | $\eta_{\mathcal{Y}}(f(x)) = L_f(\eta_{\mathcal{X}}(x))$                                           | $\eta_{\mathcal{Y}}(f(x)) = L_f(\eta_{\mathcal{X}}(x))$ , which reads $ f(x)\rangle = L_f x\rangle$ , a com-                                                                 |
| rph8-g15q_F00466 | 22   | ■ 1.000 | $ f(x)\rangle=L_f x\rangle$                                                                            | $ f(x)\rangle = L_f x\rangle$                                                                     | $\eta_{\mathcal{Y}}(f(x)) = L_f(\eta_{\mathcal{X}}(x))$ , which reads $ f(x)\rangle = L_f x\rangle$ , a com-                                                                 |
| rph8-g15q_F00467 | 22   | ■ 1.000 | $\mathbb{B}G$                                                                                          | $\mathbb{B}G$                                                                                     | group and let $\mathbb{B}G$ be the category consisting of only a sin-                                                                                                        |
| rph8-g15q_F00468 | 22   | ■ 1.000 | $\alpha: \mathbb{B}G \rightarrow$                                                                      | $\alpha : \mathbb{B}G \rightarrow$                                                                | category. A $G$ -set can then be viewed as a functor $\alpha : \mathbb{B}G \rightarrow$                                                                                      |
| rph8-g15q_F00469 | 22   | ■ 1.000 | $g^{-1}$                                                                                               | $g^{-1}$                                                                                          | invertible, there exists a $g^{-1}$ such that $gg^{-1} = 1_G = g^{-1}g$ ,                                                                                                    |
| rph8-g15q_F00470 | 22   | ■ 1.000 | $gg^{-1}=1_G=g^{-1}g$                                                                                  | $gg^{-1} = 1_G = g^{-1}g$                                                                         | invertible, there exists a $g^{-1}$ such that $gg^{-1} = 1_G = g^{-1}g$ ,                                                                                                    |
| rph8-g15q_F00471 | 22   | ■ 0.987 | $\mathrm{id}_{\mathcal{X}}=\alpha_e=\alpha_{gg^{-1}}=\alpha_g \circ \alpha_{g^{-1}}$                   | $\mathrm{id}_{\mathcal{X}} = \alpha_e = \alpha_{gg^{-1}} = \alpha_g \circ \alpha_{g^{-1}}$        | of $\alpha$ , we have that $\mathrm{id}_{\mathcal{X}} = \alpha_e = \alpha_{gg^{-1}} = \alpha_g \circ \alpha_{g^{-1}}$ , and simi-                                            |
| rph8-g15q_F00472 | 22   | ■ 0.693 | $\mathrm{id}_{\mathcal{X}}=\alpha_{g^{-1}} \circ \alpha_g$                                             | $\mathrm{id}_{\mathcal{X}} = \alpha_{g^{-1}} \circ \alpha_g$                                      | larly $\mathrm{id}_{\mathcal{X}} = \alpha_{g^{-1}} \circ \alpha_g$ , which proves that $\alpha_g$ is a bijection with                                                        |
| rph8-g15q_F00473 | 22   | ■ 0.693 | $\alpha_g$                                                                                             | $\alpha_g$                                                                                        | larly $\mathrm{id}_{\mathcal{X}} = \alpha_{g^{-1}} \circ \alpha_g$ , which proves that $\alpha_g$ is a bijection with                                                        |
| rph8-g15q_F00474 | 22   | ■ 1.000 | $\alpha_{g^{-1}}$                                                                                      | $\alpha_{g^{-1}}$                                                                                 | inverse $\alpha_{g^{-1}}$ .                                                                                                                                                  |
| rph8-g15q_F00475 | 22   | ■ 0.986 | $\alpha, \beta$                                                                                        | $\alpha, \beta$                                                                                   | Now, consider <i>two</i> such $G$ -sets, viewed as functors $\alpha, \beta :$                                                                                                |
| rph8-g15q_F00476 | 22   | ■ 0.939 | $\mathbb{B}G \rightarrow$                                                                              | $\mathbb{B}G \rightarrow$                                                                         | $\mathbb{B}G \rightarrow \mathbf{Set}$ , where $\mathcal{X} := \alpha(\bullet)$ and $\mathcal{Y} := \beta(\bullet)$ . A natural trans-                                       |
| rph8-g15q_F00477 | 22   | ■ 0.939 | $\mathcal{X}:=\alpha(\bullet)$                                                                         | $\mathcal{X} := \alpha(\bullet)$                                                                  | $\mathbb{B}G \rightarrow \mathbf{Set}$ , where $\mathcal{X} := \alpha(\bullet)$ and $\mathcal{Y} := \beta(\bullet)$ . A natural trans-                                       |
| rph8-g15q_F00478 | 22   | ■ 0.939 | $\mathcal{Y}:=\beta(\bullet)$                                                                          | $\mathcal{Y} := \beta(\bullet)$                                                                   | $\mathbb{B}G \rightarrow \mathbf{Set}$ , where $\mathcal{X} := \alpha(\bullet)$ and $\mathcal{Y} := \beta(\bullet)$ . A natural trans-                                       |
| rph8-g15q_F00479 | 22   | ■ 0.949 | $f: \alpha \Rightarrow \beta$                                                                          | $f : \alpha \Rightarrow \beta$                                                                    | formation $f : \alpha \Rightarrow \beta$ assigns to the only object $\bullet$ of $\mathbb{B}G$ a                                                                             |
| rph8-g15q_F00480 | 22   | ■ 0.949 | $\bullet$                                                                                              | $\bullet$                                                                                         | formation $f : \alpha \Rightarrow \beta$ assigns to the only object $\bullet$ of $\mathbb{B}G$ a                                                                             |
| rph8-g15q_F00481 | 25   | ■ 1.000 | $w^*$                                                                                                  | $w^*$                                                                                             | uct measures to the tensor product of semifinite $w^*$ -algebras,                                                                                                            |
| rph8-g15q_F00482 | 25   | ■ 1.000 | $x, y, \ldots \in \mathbb{R}$                                                                          | $x, y, \dots \in \mathbb{R}$                                                                      | whose objects are elements $x, y, \dots \in \mathbb{R}$ and where there is                                                                                                   |
| rph8-g15q_F00483 | 25   | ■ 1.000 | $x \rightarrow y$                                                                                      | $x \rightarrow y$                                                                                 | a morphism $x \rightarrow y$ whenever $x \leqslant y$ . This is a category due                                                                                               |
| rph8-g15q_F00484 | 25   | ■ 1.000 | $x \leqslant y$                                                                                        | $x \leqslant y$                                                                                   | a morphism $x \rightarrow y$ whenever $x \leqslant y$ . This is a category due                                                                                               |
| rph8-g15q_F00485 | 25   | ■ 1.000 | $\alpha \in \mathbb{R}^2$                                                                              | $\alpha \in \mathbb{R}^2$                                                                         | in category theory. The vector $\alpha \in \mathbb{R}^2$ is identified with the                                                                                              |
| rph8-g15q_F00486 | 25   | ■ 0.999 | $\alpha: \mathbb{R} \rightarrow \mathbb{R}^2$                                                          | $\alpha : \mathbb{R} \rightarrow \mathbb{R}^2$                                                    | additive map $\alpha : \mathbb{R} \rightarrow \mathbb{R}^2$ sending $1 \in \mathbb{R}$ to $\alpha$ .                                                                         |

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| rph8-g15q_F00487 | 25   | ■ 0.999 | <code>1 \in \mathbb{R}</code>                                                         | $1 \in \mathbb{R}$                                                          | additive map $\alpha : \mathbb{R} \rightarrow \mathbb{R}^2$ sending $1 \in \mathbb{R}$ to $\alpha$ .                              |
| rph8-g15q_F00488 | 25   | ■ 1.000 | <code>\operatorname{Tr}[\sigma\,0]</code>                                             | $\operatorname{Tr}[\sigma O]$                                               | $\operatorname{Tr}[\sigma O]$ defines a hyperplane $\{\sigma \in \mathcal{S}(\mathcal{H}) : \operatorname{Tr}[\sigma O] = 0\}$ in |
| rph8-g15q_F00489 | 25   | ■ 1.000 | <code>\{\sigma \in \mathscr{S}(\mathcal{H}) : \operatorname{Tr}[\sigma\,0]=0\}</code> | $\{\sigma \in \mathcal{S}(\mathcal{H}) : \operatorname{Tr}[\sigma O] = 0\}$ | $\operatorname{Tr}[\sigma O]$ defines a hyperplane $\{\sigma \in \mathcal{S}(\mathcal{H}) : \operatorname{Tr}[\sigma O] = 0\}$ in |